Neurosymbolic Knowledge Bases & World Modeling

Lara J. Martin (she/they)

https://laramartin.net/neurosymbolic-text-gen/

Learning Objectives

- Follow examples of finetuning and post-hoc knowledge base integration
- Consider how knowledge bases might still be useful with today's LLMs
- Find the connection between world modeling, state tracking, and experience managers
- Evaluate how world modeling has evolved through the years

Review: Desirable properties for a commonsense resource

COVERAGE

Large scale

Diverse knowledge types

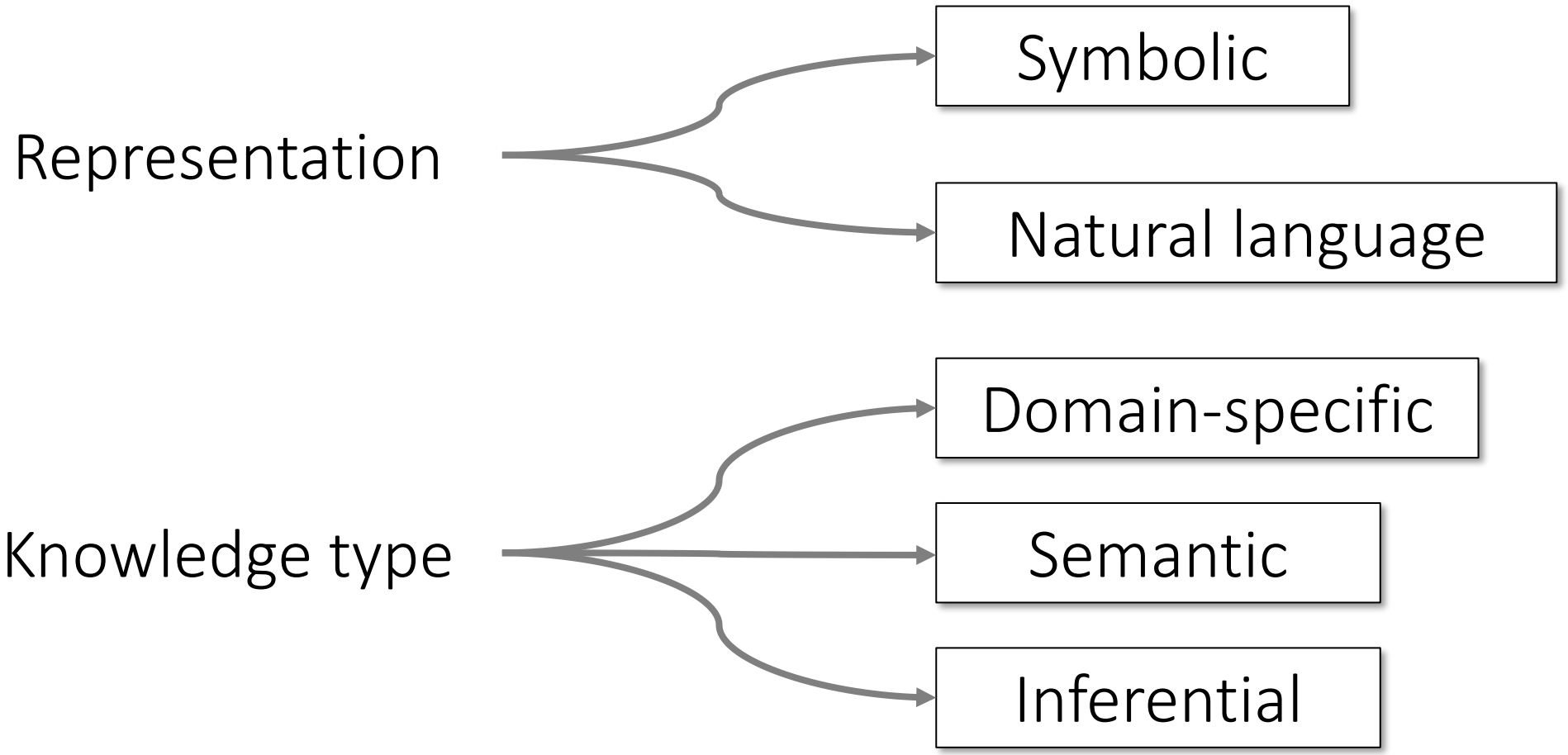
USEFUL

High quality knowledge

Usable in downstream tasks

Multiple resources tackle different knowledge types

Review: Creating a commonsense resource



Review: Some commonsense cannot be extracted

Text is subject to **reporting bias**
(Gordon & Van Durme, 2013)

Noteworthy events

- Murdering 4x more common than exhaling

Commonsense is not often written

- Grice's maxim of quantity



found when extracting commonsense knowledge on four large corpora using Knext (Gordon & Van Durme, 2013)

When communicating, people try to be as informative as they possibly can, and give as much information as is needed, and no more.

Neurosymbolic Knowledge Bases

Types of Neurosymbolic KBs

Like other neurosymbolic systems, the symbolic component can be integrated into the neural component in various ways

We are going to look at two example systems:

- One that trains on the KB
- One that uses the KB as a “wrapper”

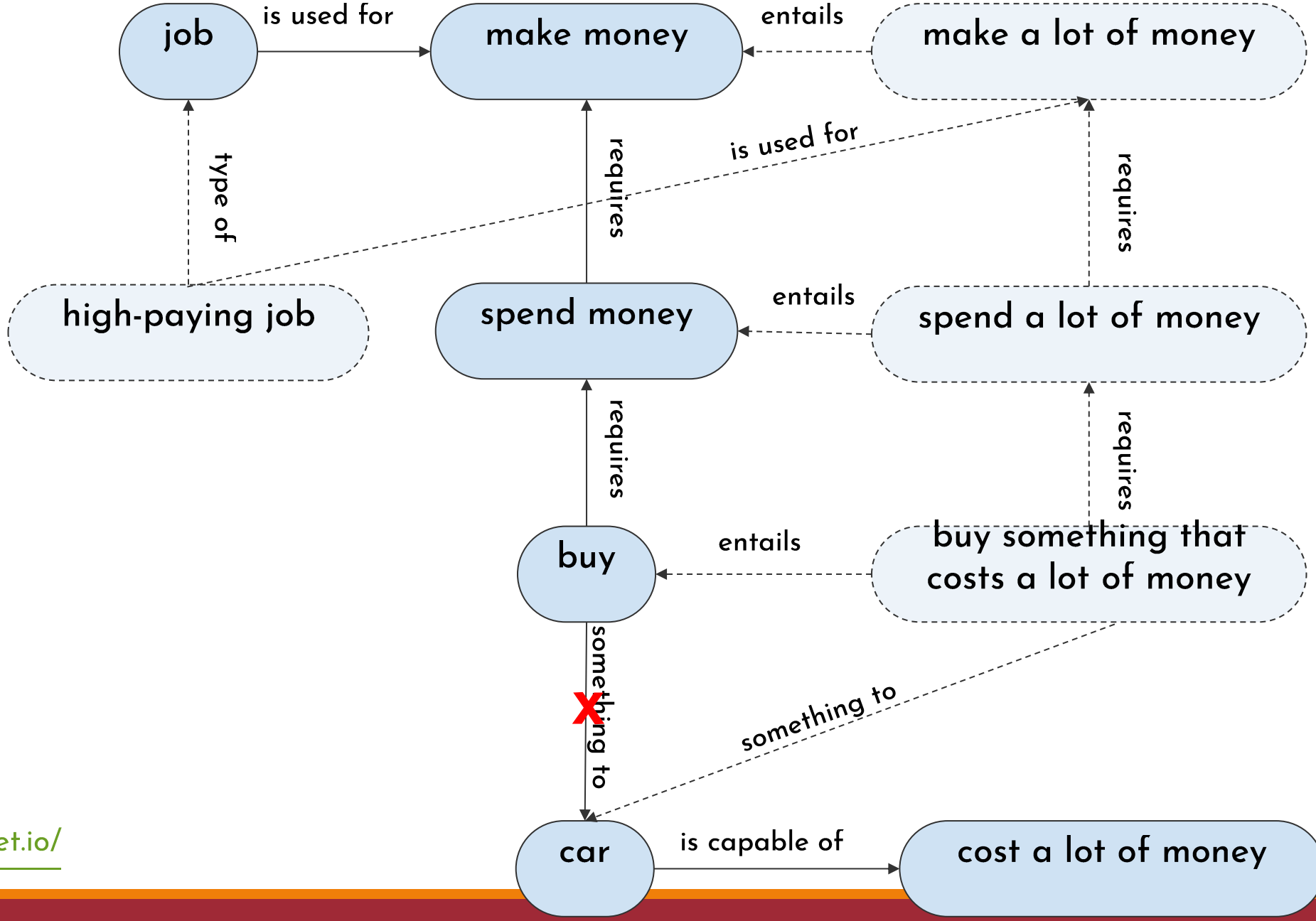
BERT would struggle with commonsense

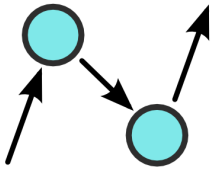
[CLS] Katrina had the financial means to afford a new car while Monica did not, since [SEP] Katrina had a high paying job.

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0.51
0.49

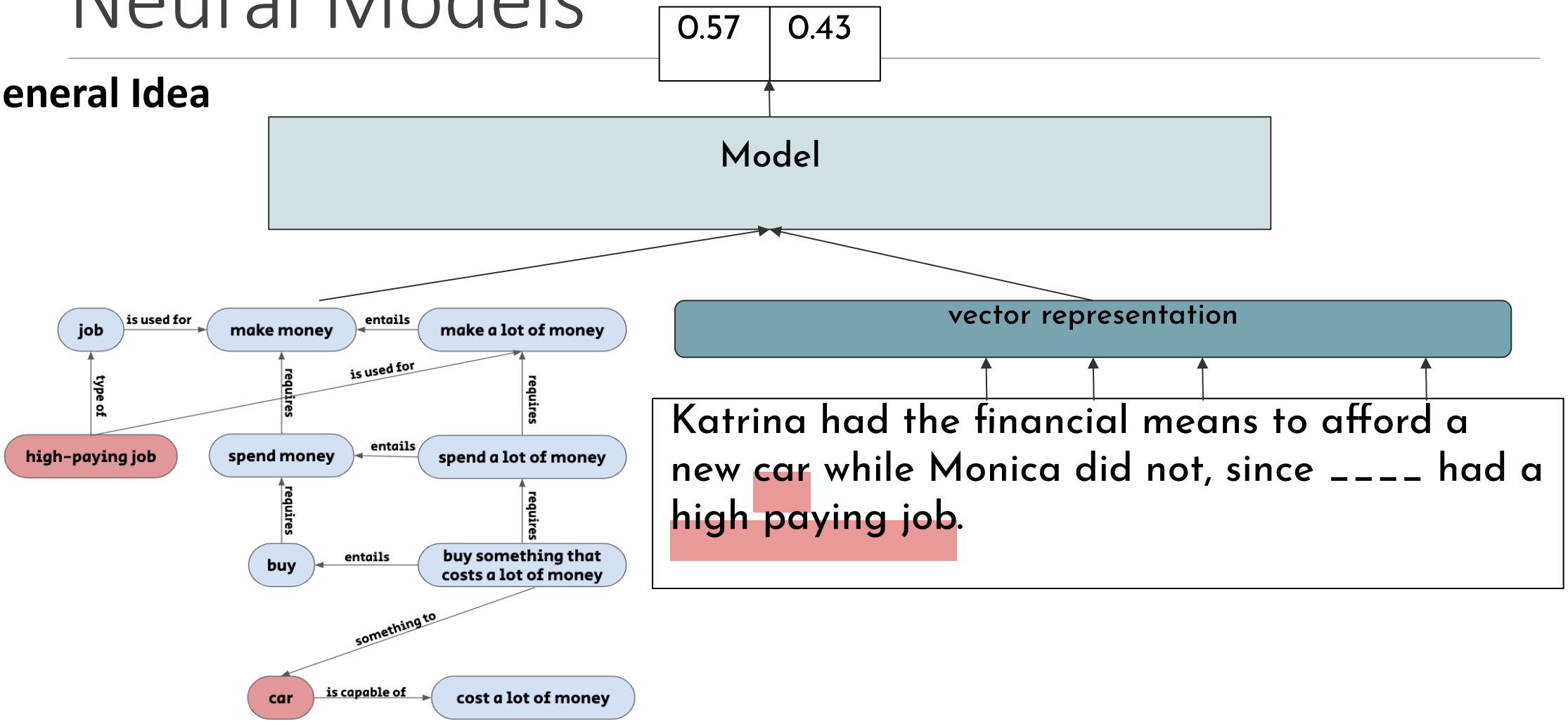




https://conceptnet.io/

Incorporating External Knowledge into Neural Models

General Idea

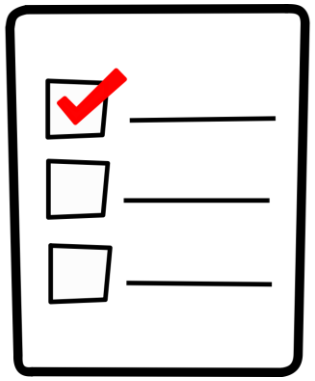


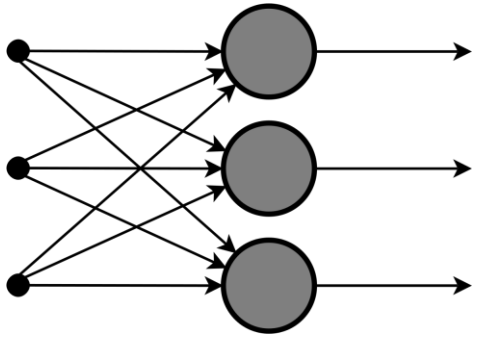
Incorporating External Knowledge into Neural Models

Recipe

Task

Story ending,
Machine Comprehension
Social common sense
NLI





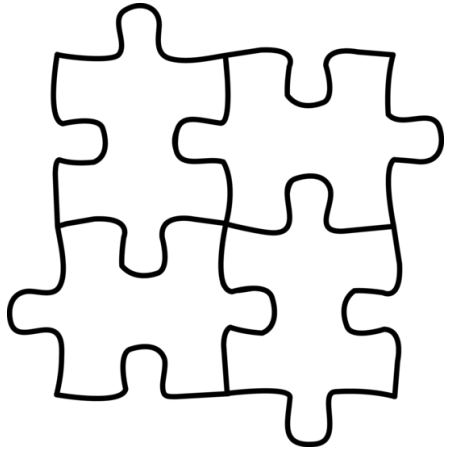
Neural Component

Pre/post pre-trained language models

Knowledge Source

Knowledge bases, extracted from text, hand-crafted rules

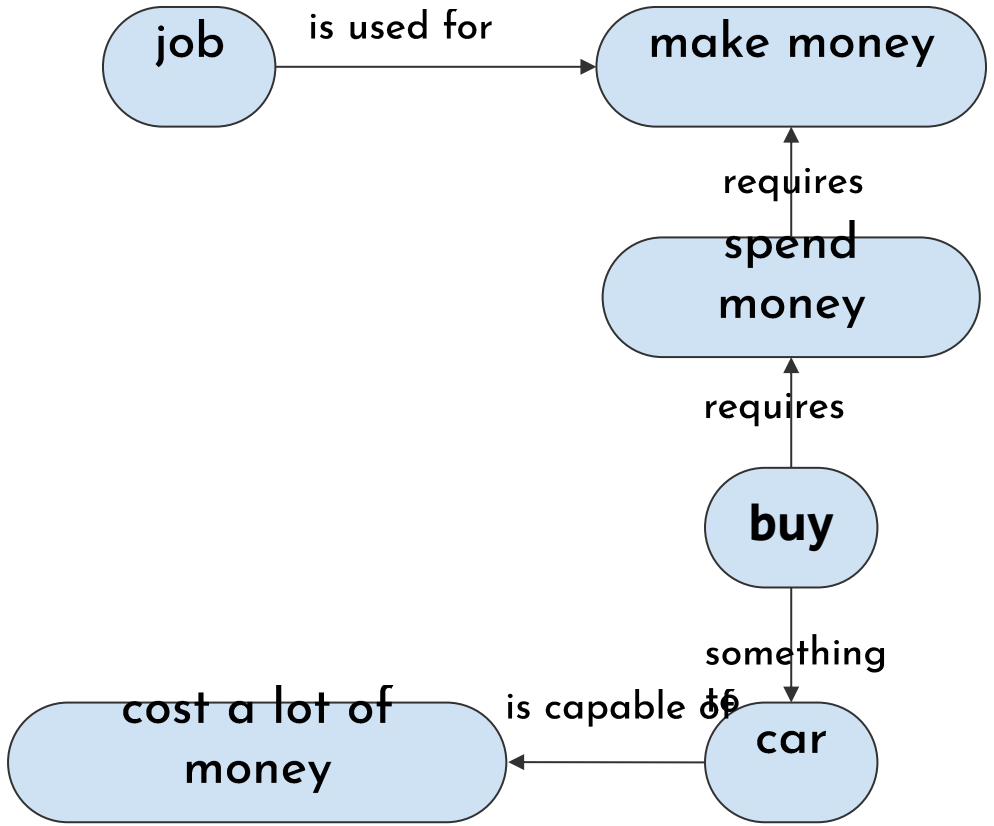




Combination Method

Attention, pruning, word embeddings, multi-task learning

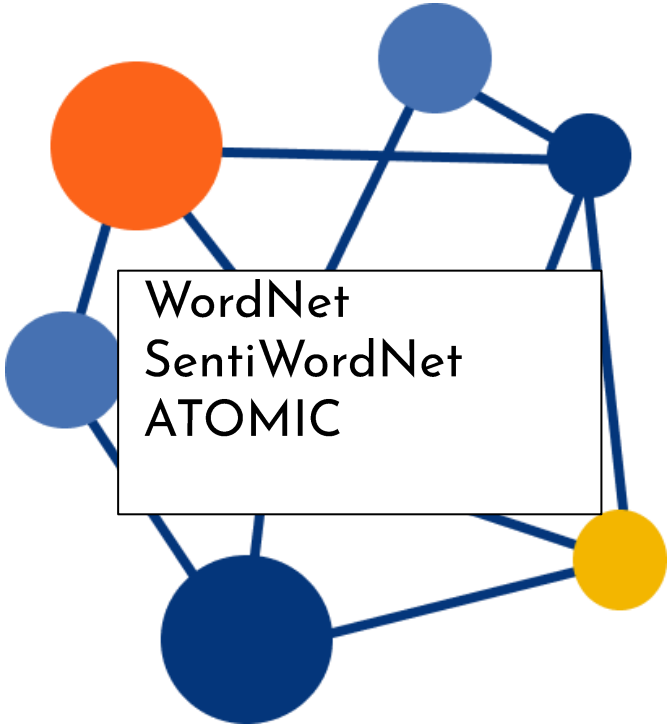
ConceptNet



Knowledge Source

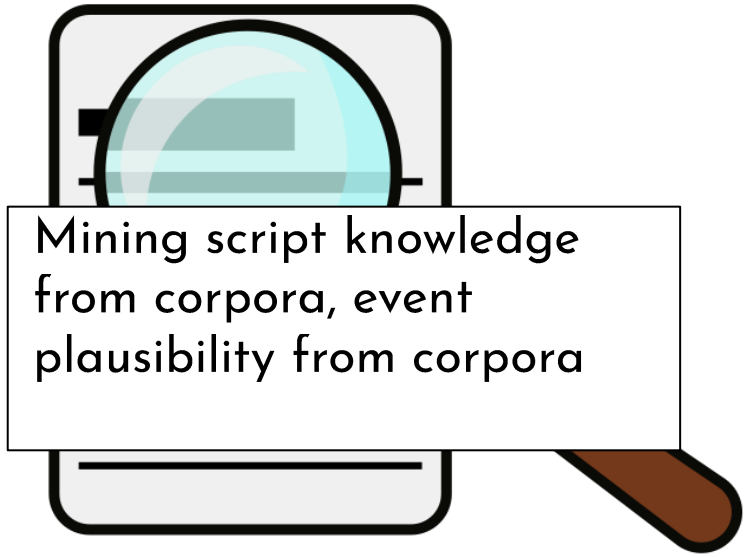
Conceptnet 5.5: An open multilingual graph of general knowledge. *Robyn Speer, Joshua Chin, and Catherine Havasi.* AAAI 2017.

Other Knowledge Sources



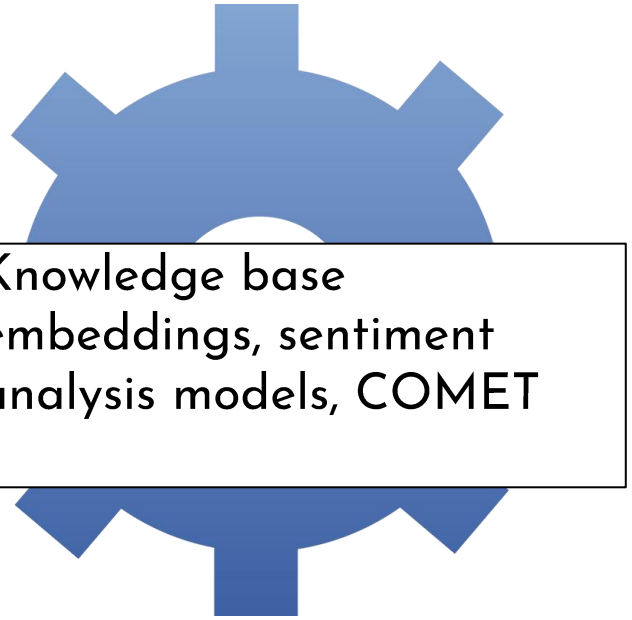
WordNet
SentiWordNet
ATOMIC

Knowledge Bases



Mining script knowledge from corpora, event plausibility from corpora

Mining from Text



Knowledge base embeddings, sentiment analysis models, COMET

Tools

Neural Component

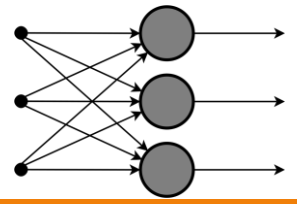
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[CLS] Katrina had the financial means to afford a new car while Monica did not, since [SEP] Monica had a high paying job.



0.51
0.49

Neural



COMET's Combination Method

Incorporate into scoring function

Symbolic → vector representation

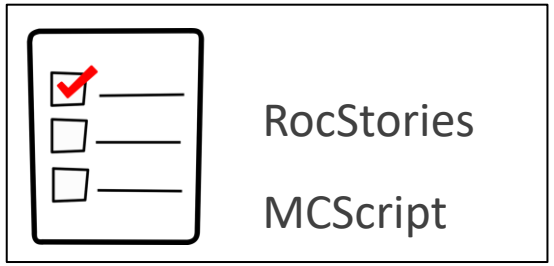
Multi-task learning

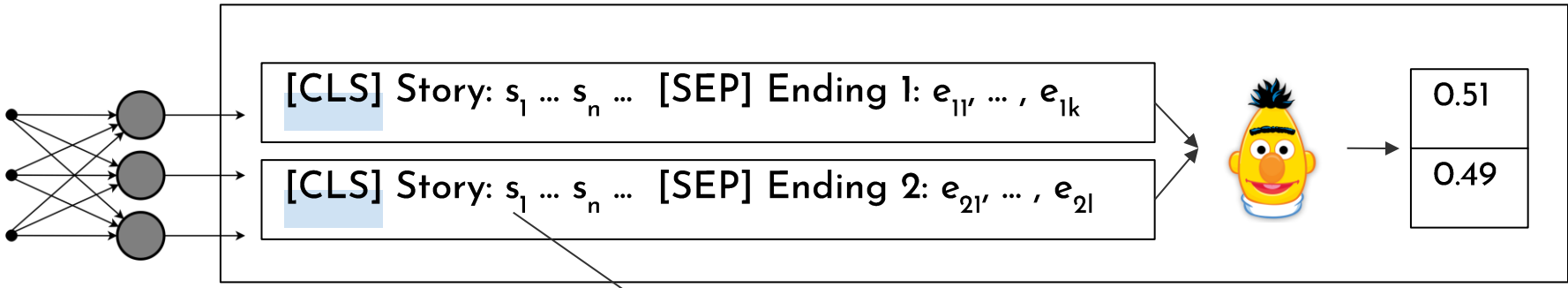
- (This was before we had *very large* LMs)

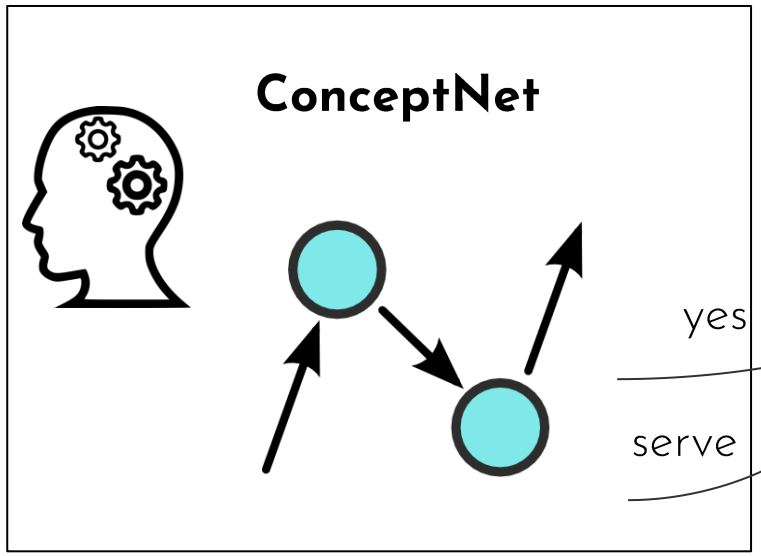


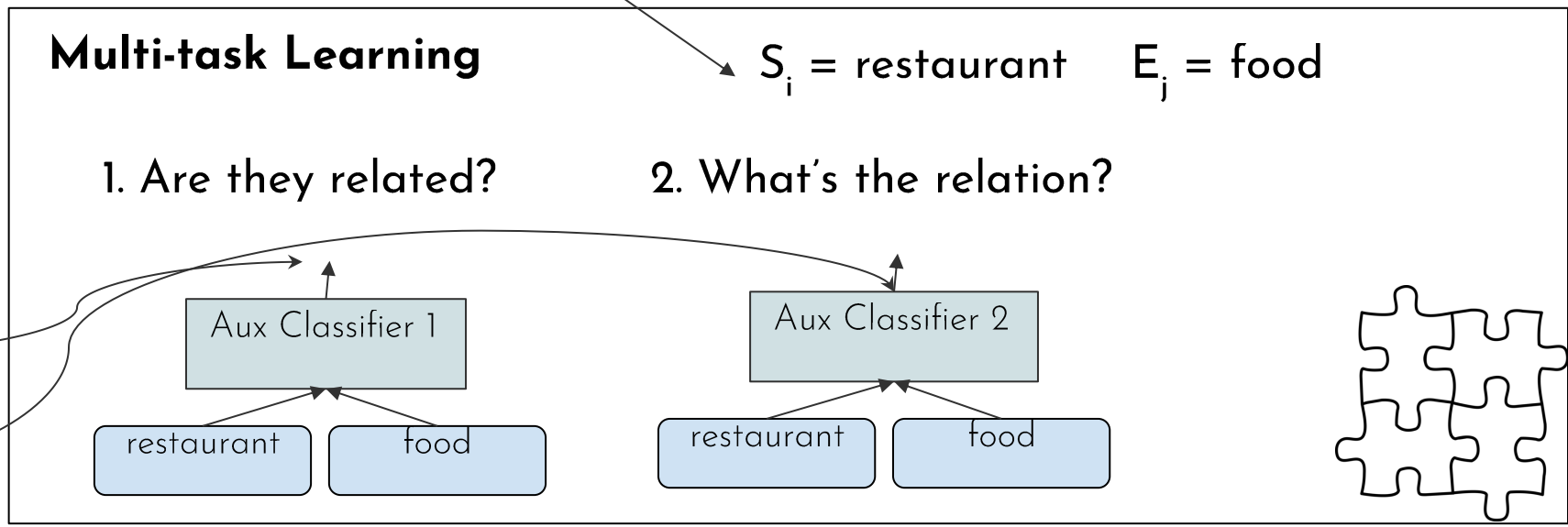
Incorporating External Knowledge into Neural Models

Example









Incorporating Commonsense Reading Comprehension with Multi-task Learning. *Jiangnan Xia, Chen Wu, and Ming Yan.* CIKM 2019.

Constructing Knowledge Graphs

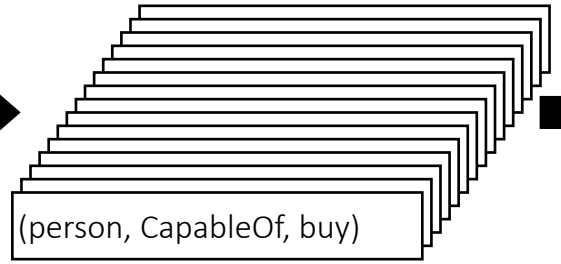
Observe world

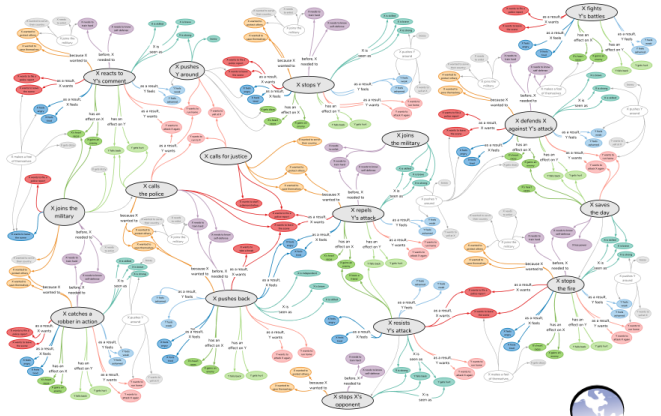


Write commonsense knowledge facts



Store facts in knowledge graph





(Miller, 1995)

(Singh et al., 2002)

(Lenat, 1995)

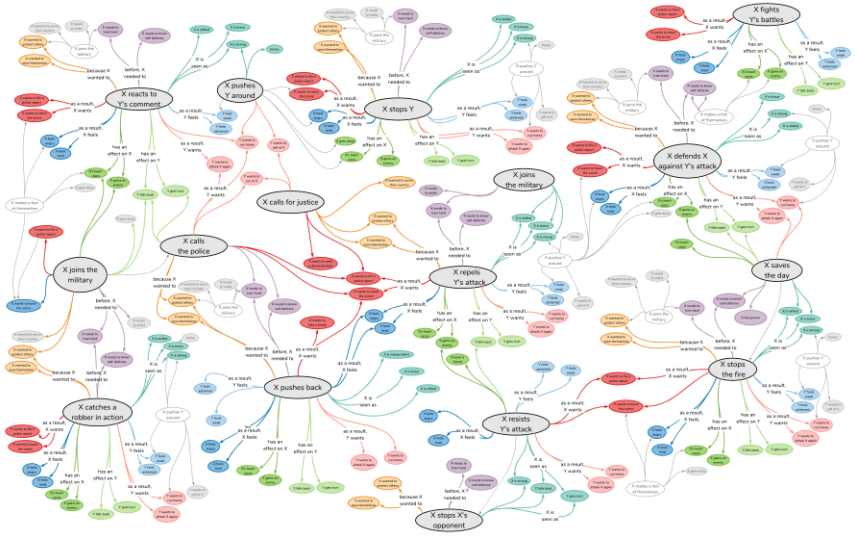


(Sap et al., 2019)

Limitations of Knowledge Graphs

- Situations rarely found as-is in commonsense knowledge graphs

ATOMIC



(Sap et al., 2019)

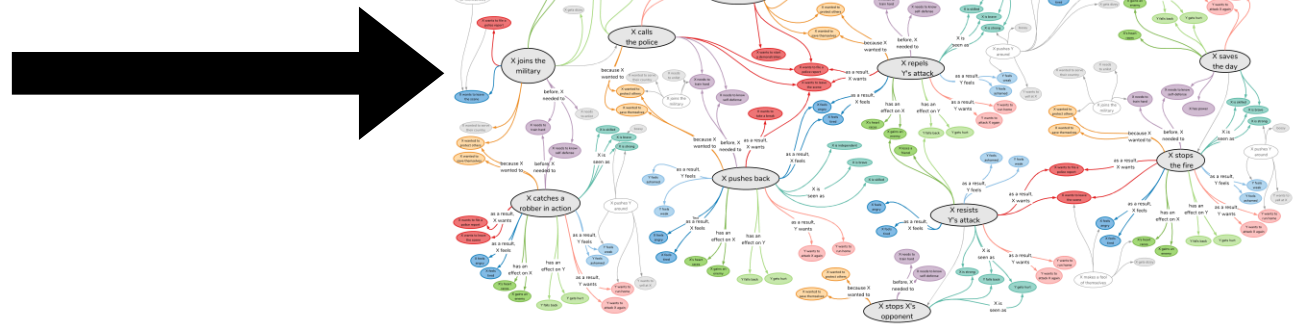
(X goes to the mall, Effect on X, buys clothes)

(X goes the mall, Perception of X, rich)

(X gives Y some money, Reaction of Y, grateful)

Limitations of Knowledge Graphs

Kai knew that things were getting out of control and managed to keep his temper in check

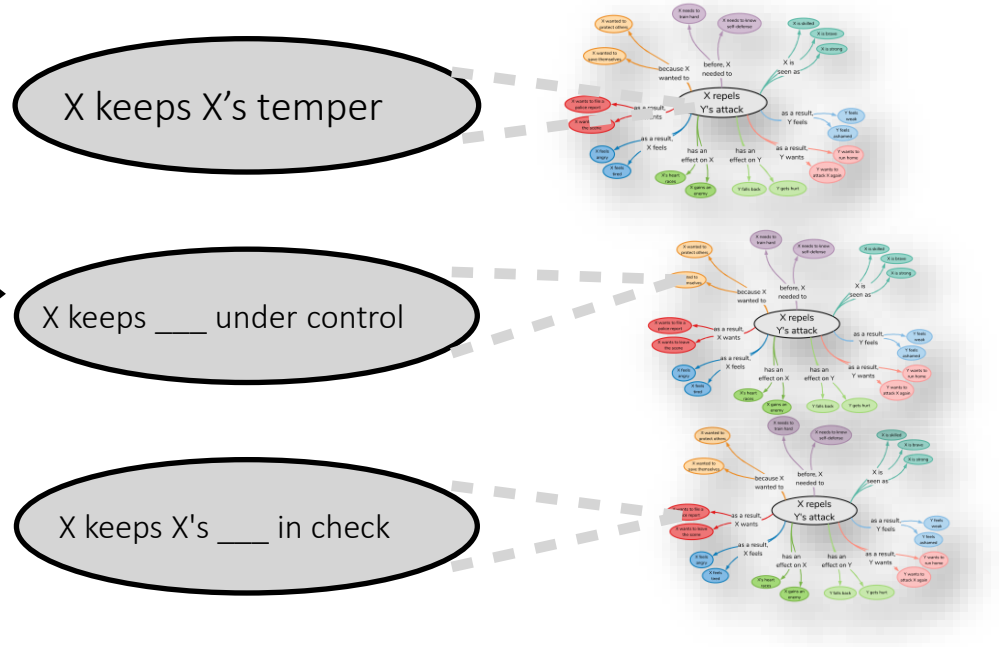


(Sap et al., 2019)

Limitations of Knowledge Graphs

- Situations rarely found as-is in commonsense knowledge graphs
- Connecting to knowledge graphs can yield incorrect nodes

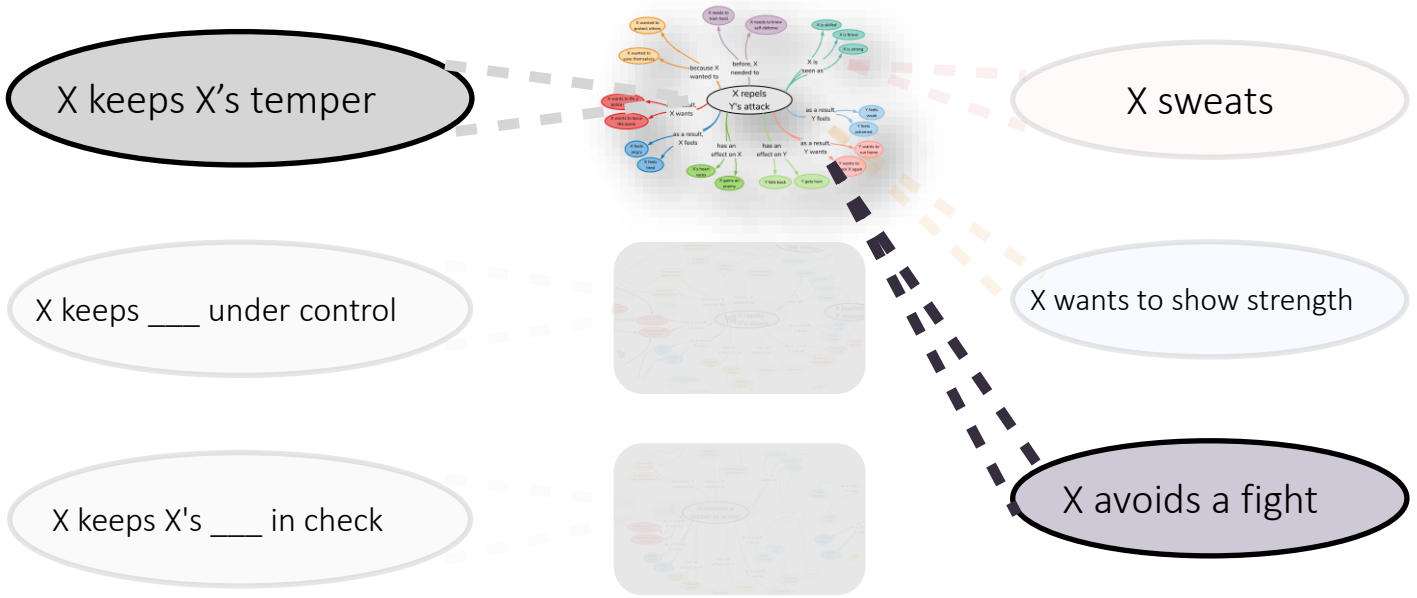
Kai knew that things were getting out of control and managed to keep his temper in check



Limitations of Knowledge Graphs

- Situations rarely found as-is in commonsense knowledge graphs
- Connecting to knowledge graphs can yield incorrect nodes
- Suitable nodes are often uncontextualized

Kai knew that things were getting out of control and managed to keep his temper in check



Challenges of Prior Approaches

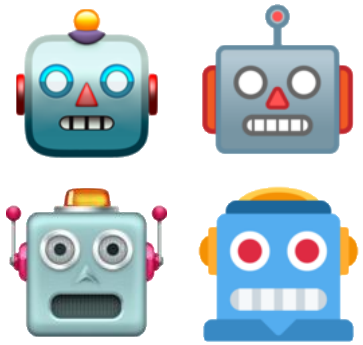
Commonsense knowledge is immeasurably vast, making it impossible to manually enumerate

Constructing Knowledge Graphs Automatically

Gather Textual Corpus

John went to the grocery **store** to buy some **steaks**. He was going to prepare **dinner** for his **daughter**'s **birthday**. She was turning 5 and would be starting elementary **school** soon.

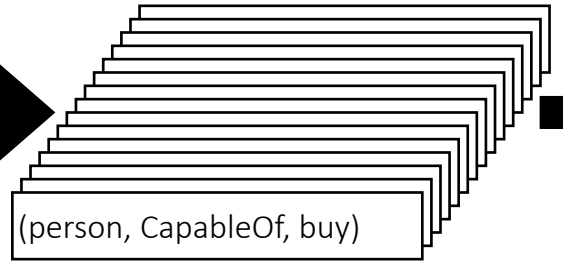
Automatically extract knowledge



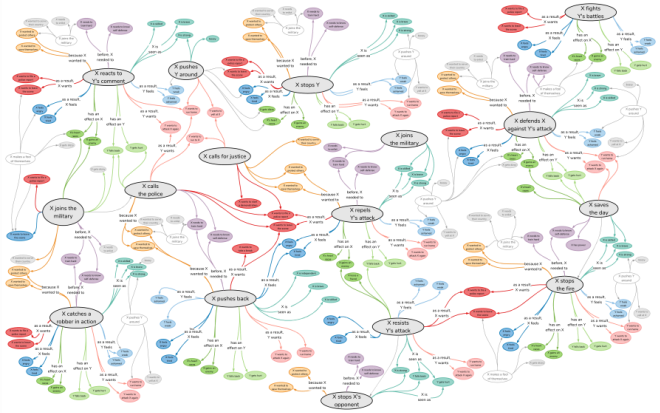
(Schubert, 2002)

(Banko et al., 2007)

(Zhang et al., 2020)



Store in knowledge graph



ConceptNet
An open, multilingual knowledge graph

(Speer et al., 2017)

Webchild

(Tandon et al., 2019)

Challenges of Prior Approaches

Commonsense knowledge is immeasurably vast, making it impossible to manually enumerate

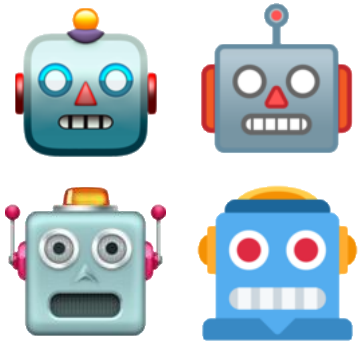
Commonsense knowledge is often implicit, and often can’t be directly extracted from text

Knowledge Base Completion

Gather training set of knowledge tuples

(person, CapableOf, buy)

Learn relationships among entities



(Socher et al., 2013)

(Bordes et al., 2013)

(Riedel et al., 2013)

(Toutanova et al., 2015)

(Yang et al., 2015)

(Trouillon et al., 2016)

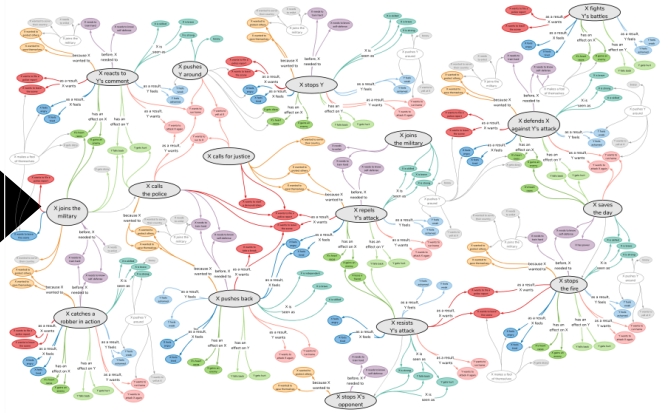
(Nguyen et al., 2016)

(Dettmers et al., 2018)

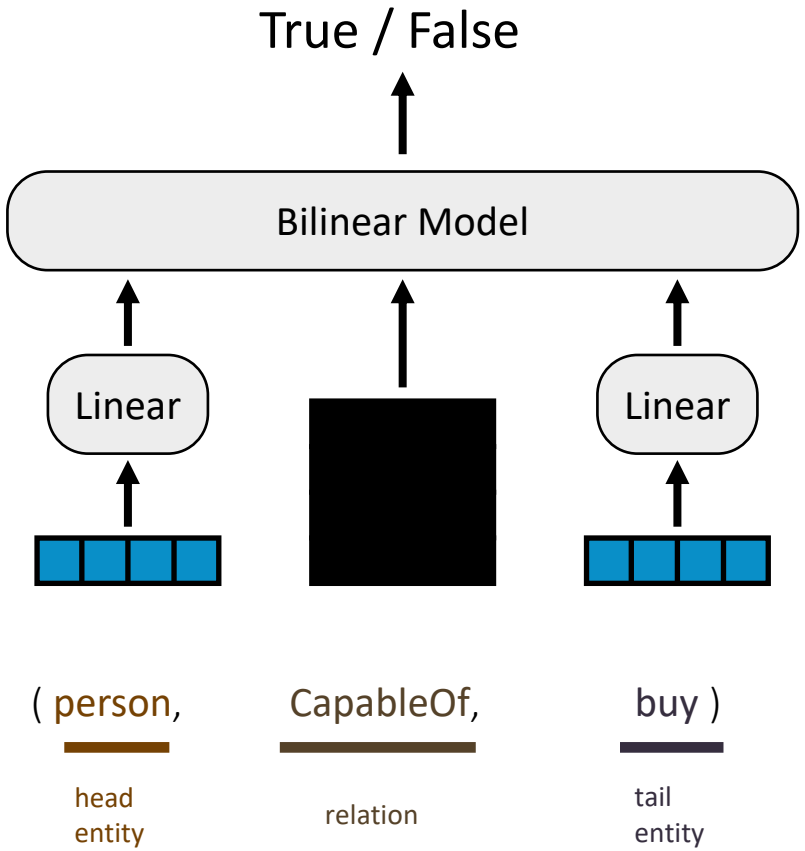
Predict new relationships

(person, CapableOf,)?

Store in knowledge graph



Commonsense Knowledge Base Completion



Li et al., 2016

Only high confidence predictions are validated

Low Novelty

Jastrzebski et al., 2018

Challenges of Prior Approaches

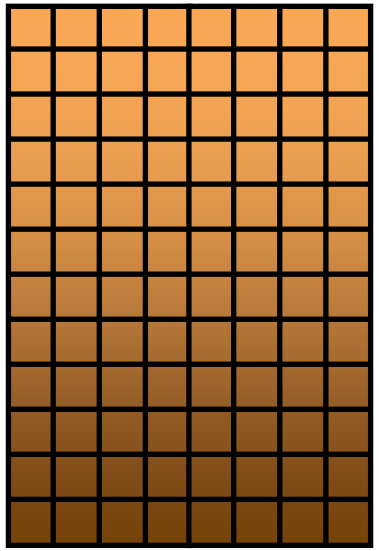
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Commonsense knowledge is often implicit, and often can’t be directly extracted from text

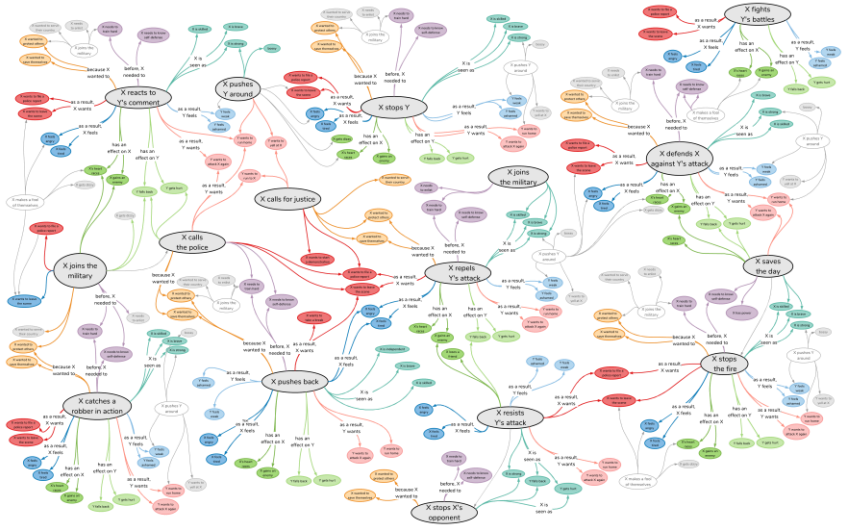
Commonsense knowledge resources are quite sparse, making them difficult to extend by only learning from examples

Solution Outline

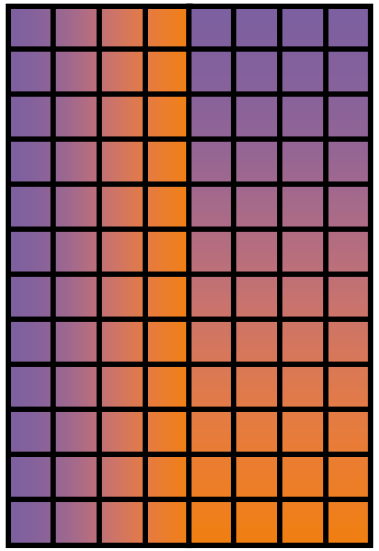
Leverage manually curated commonsense knowledge resources
Learn from the examples to induce new relationships
Scale up using language resources



Learn word embeddings from language corpus



Retrofit word embeddings on semantic resource



Learn knowledge-aware embeddings

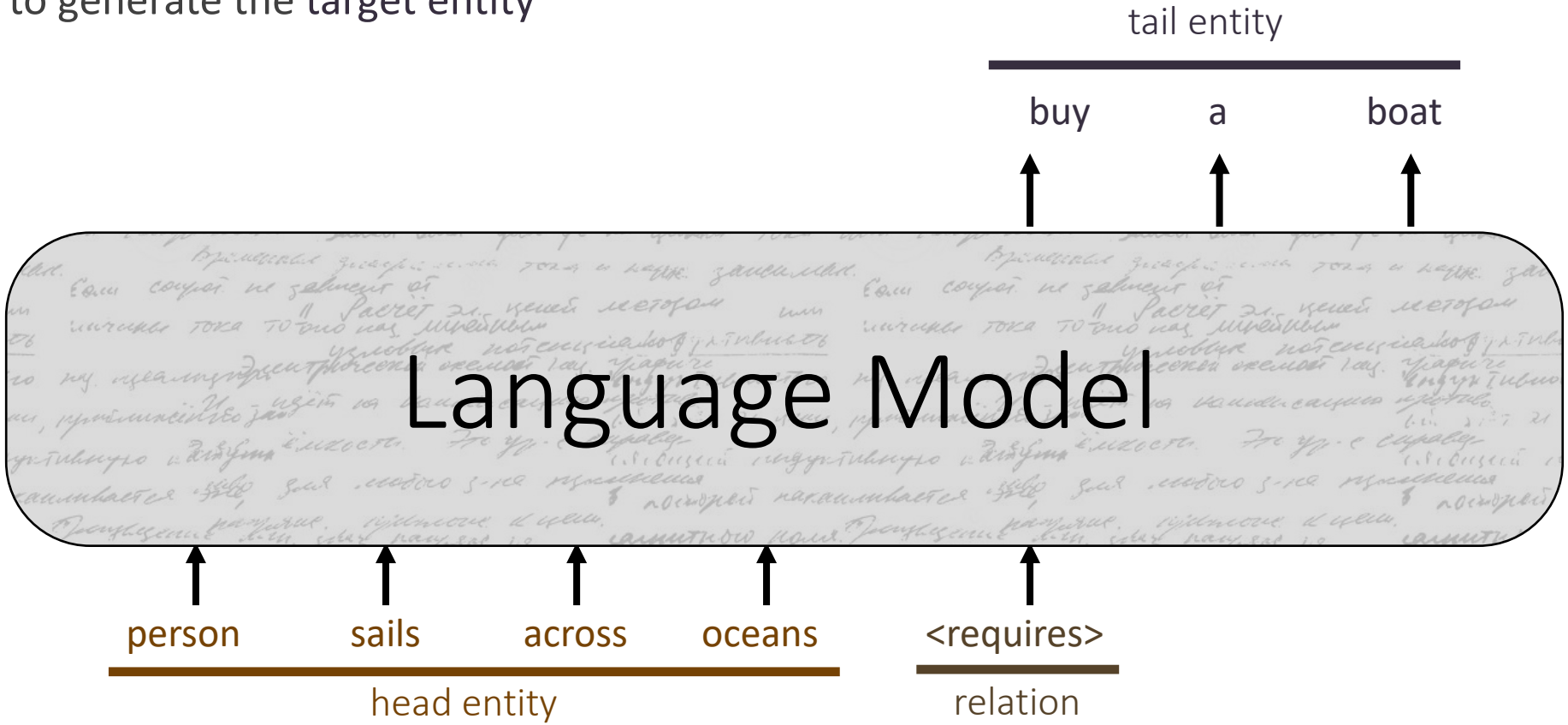
Structure of Knowledge Tuple



Learning Structure of Knowledge

Given a seed entity and a relation, learn to generate the target entity

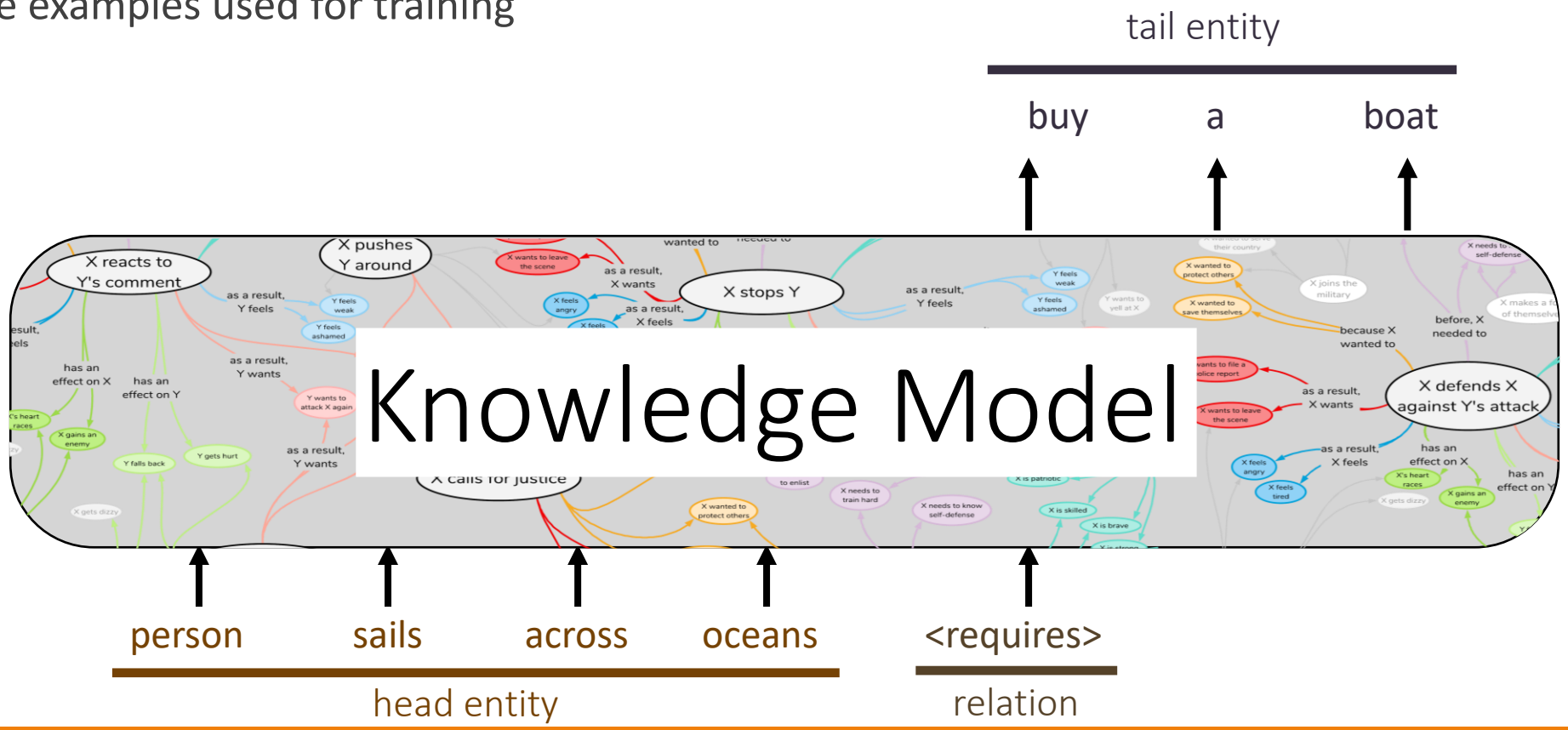
$$\mathcal{L} = -\sum \log P(\textit{target words} | \textit{seed words, relation})$$



(Bosselut et al., 2019)

Learning Structure of Knowledge

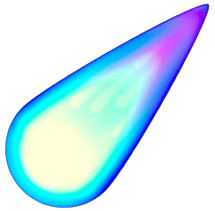
Language Model → Knowledge Model: generates knowledge of the structure of the examples used for training



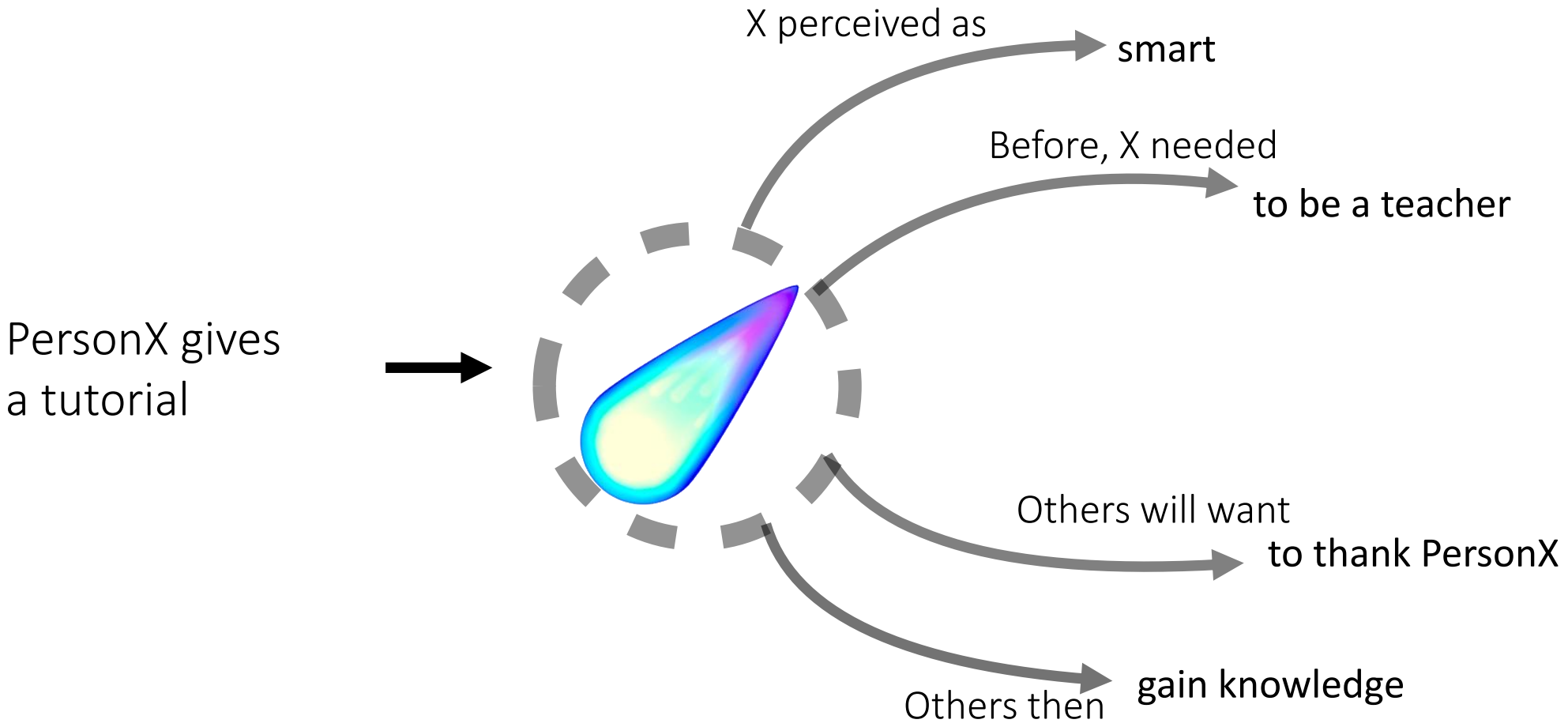
(Bosselut et al., 2019)

Generate commonsense knowledge for **any** input concept

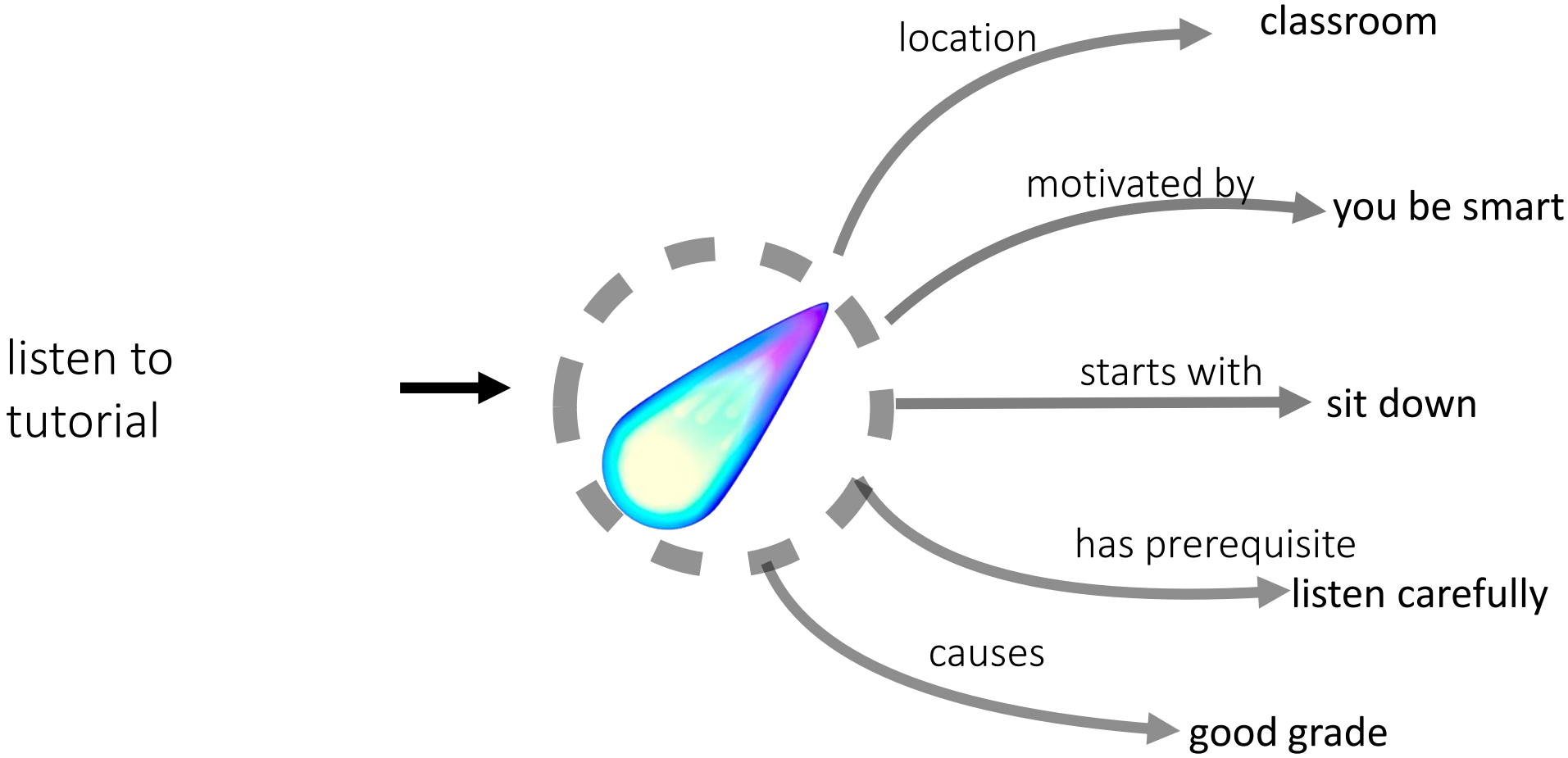
COMmons**E**nse **T**ransformers



COMET - ATOMIC

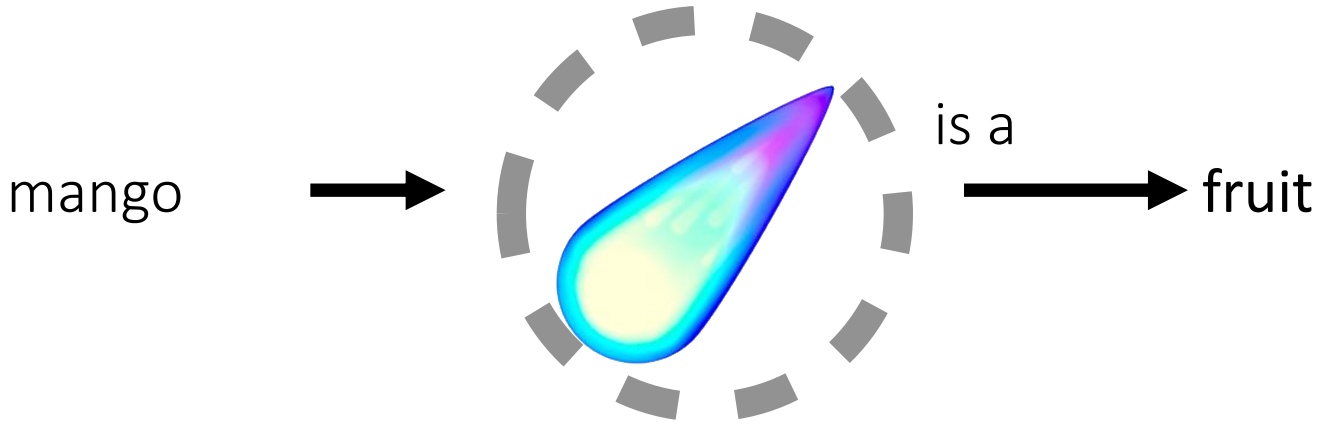


COMET - ConceptNet

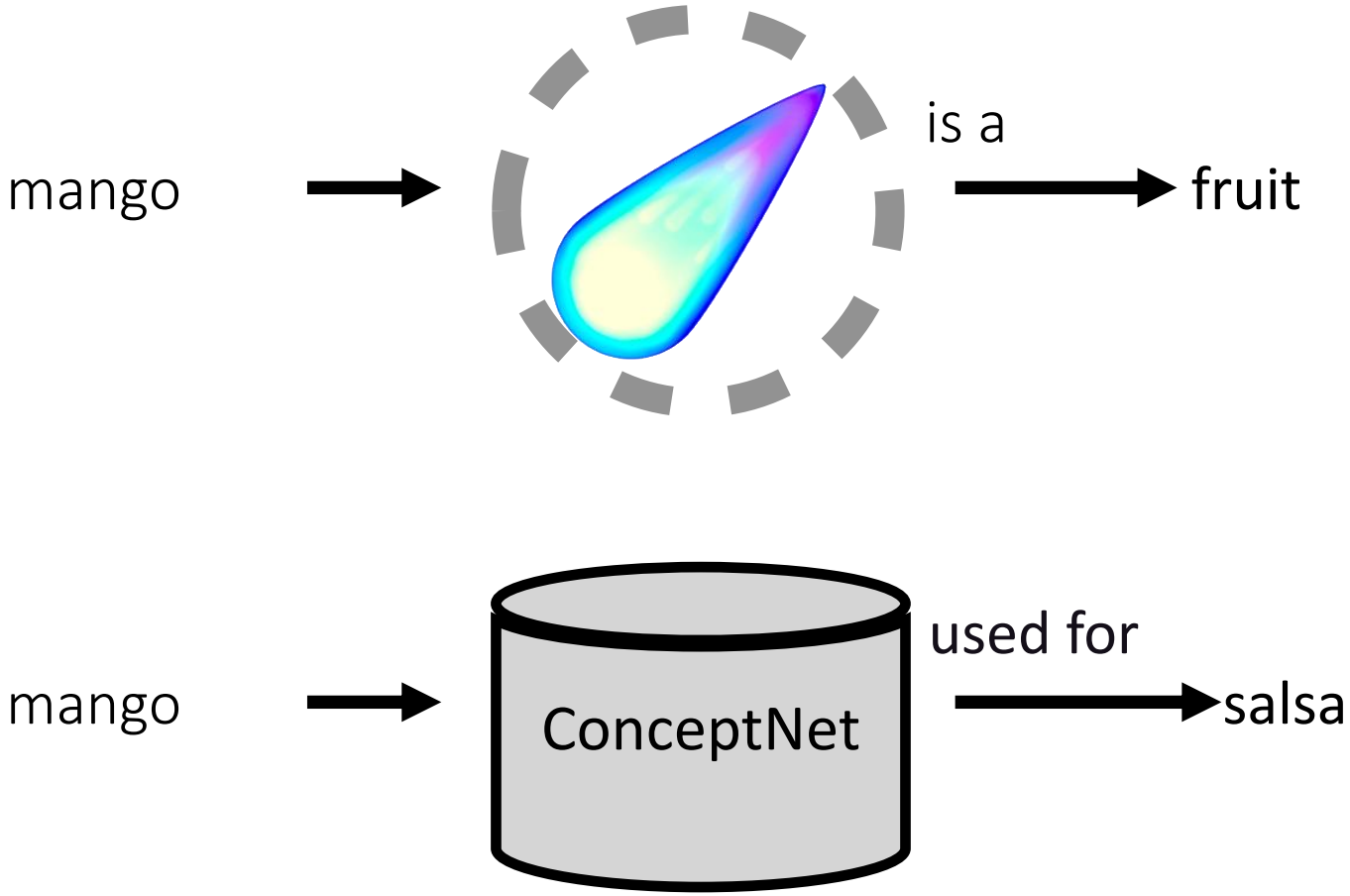


Why does this work?

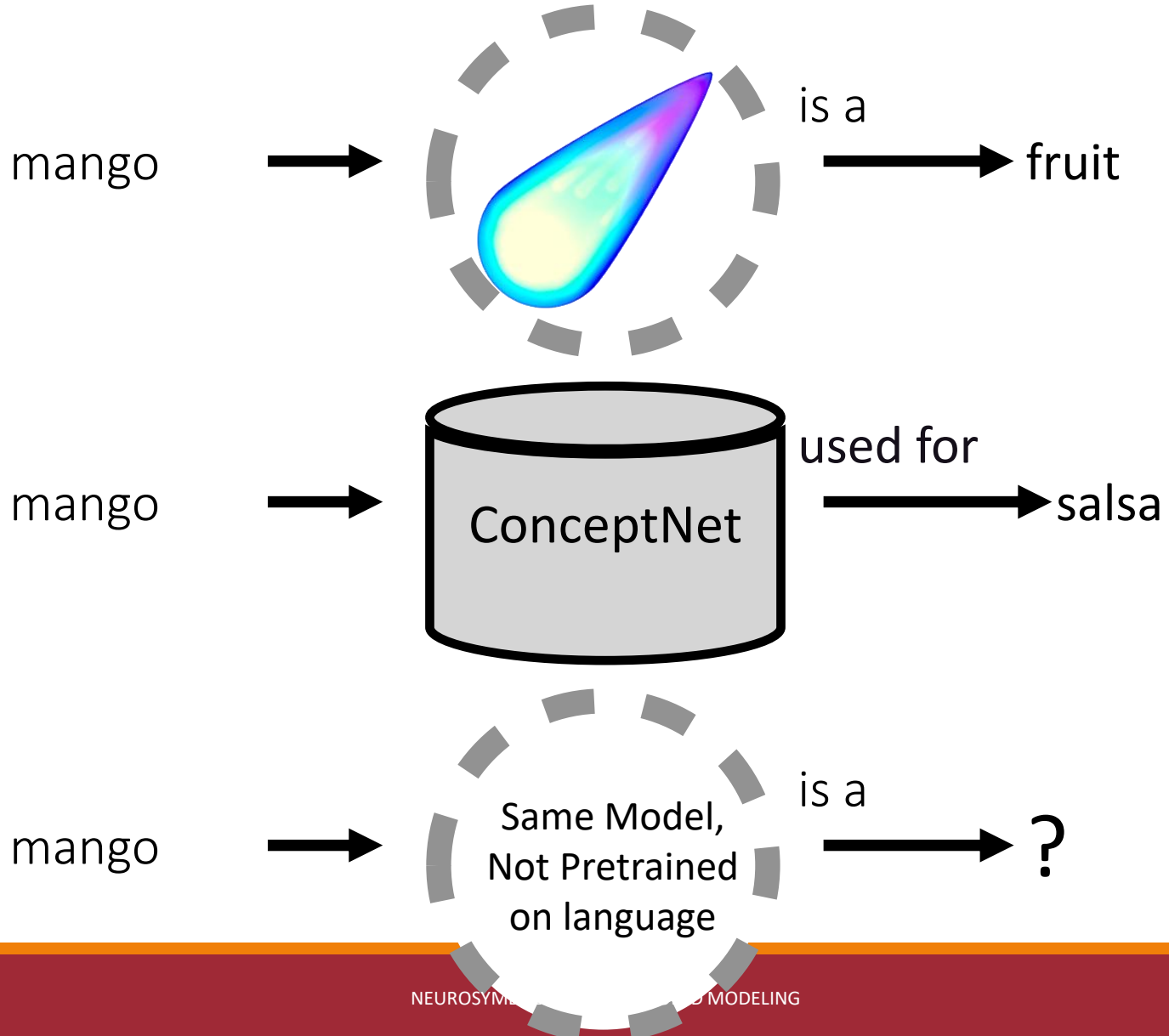
Transfer Learning from Language



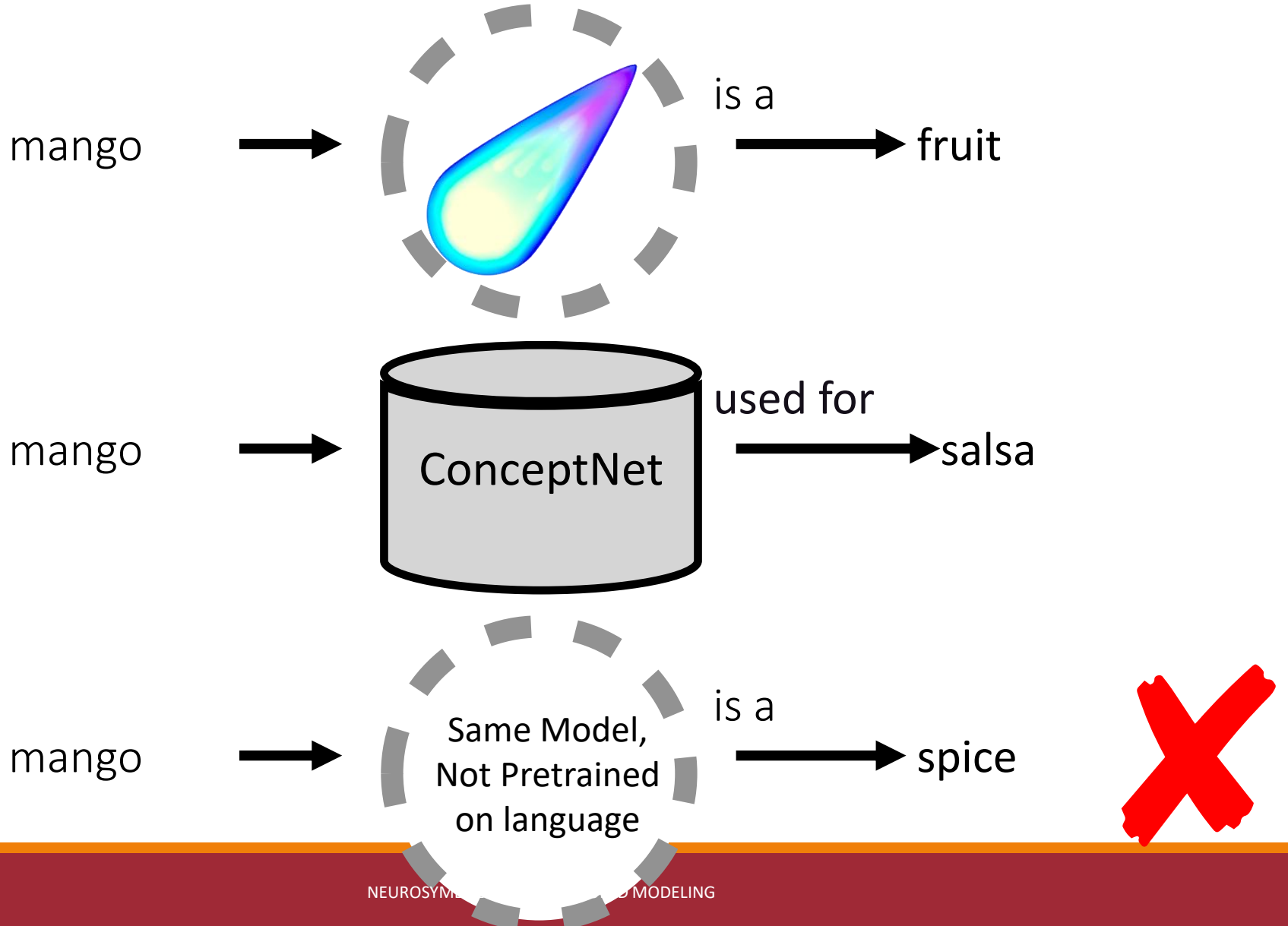
Transfer Learning from Language



Transfer Learning from Language



Transfer Learning from Language



Can an off-the-shelf language model do the same thing?

Yes, but with guidance

Did GPT-3-era Language Models know this?

Sentence:

mango is a

Predictions:

2.1% great
1.9% very
1.2% new
1.0% good
1.0% small
← Undo

Sentence:

a mango is a

Predictions:

4.2% good
4.0% very
2.5% great
2.4% delicious
1.8% sweet
← Undo

Sentence:

A mango is a

Predictions:

4.2% fruit
3.5% very
2.5% sweet
2.2% good
1.5% delicious
← Undo

Did *Masked* Language Models (like BERT) know this?

Sentence:

mango is a [MASK]

Mask 1 Predictions:

69.7% .
9.3% ;
1.7% !
0.8% vegetable
0.7% ?

Sentence:

mango is a [MASK] .

Mask 1 Predictions:

7.6% staple
7.6% vegetable
4.6% plant
3.5% tree
3.5% fruit

Sentence:

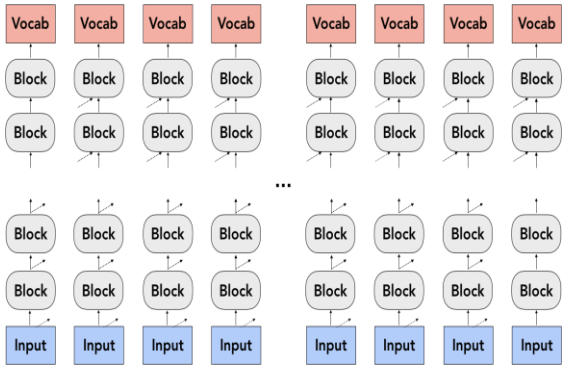
A mango is a [MASK] .

Mask 1 Predictions:

16.0% banana
12.1% fruit
5.9% plant
5.5% vegetable
2.5% candy

Commonsense Transformers

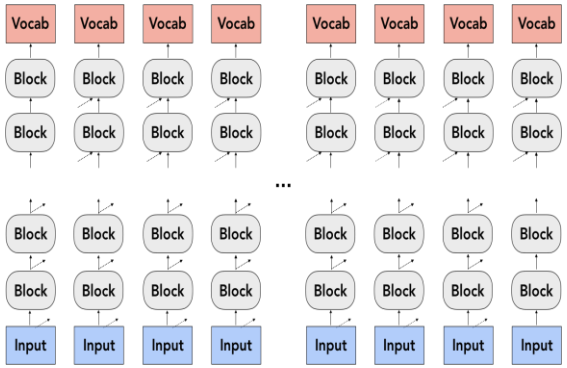
- Language models **implicitly represent knowledge**



Pre-trained
Language Model

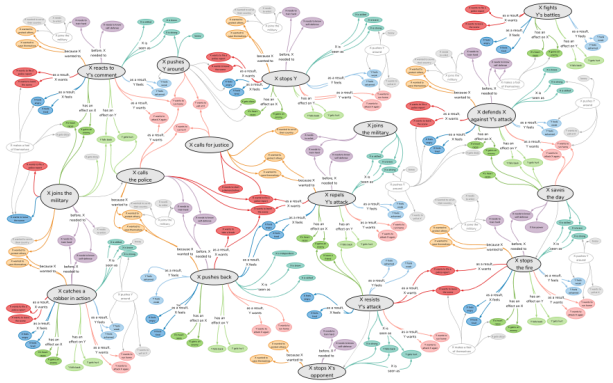
Commonsense Transformers

- Language models **implicitly represent knowledge**
- Finetune them on knowledge graphs to **learn structure of knowledge**



Pre-trained
Language Model

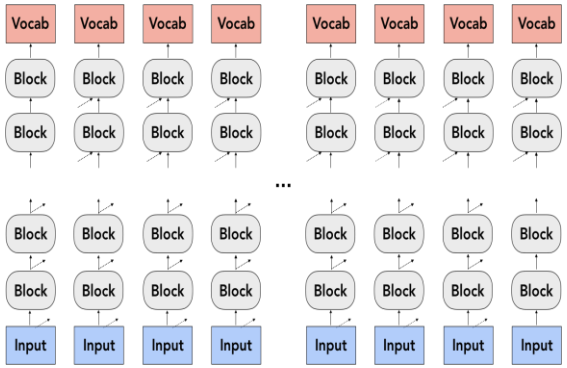
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Seed Knowledge
Graph Training

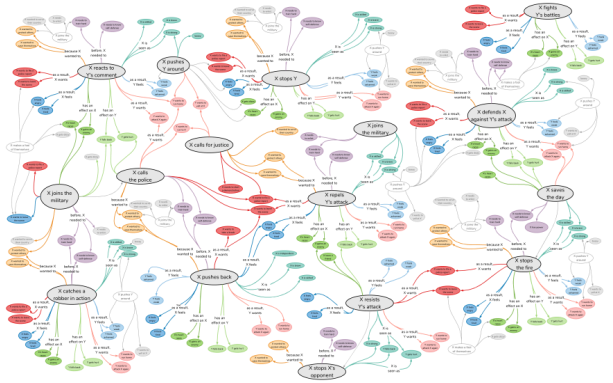
Commonsense Transformers

- Language models **implicitly represent knowledge**
- Finetune them on knowledge graphs to **learn structure of knowledge**
- Resulting knowledge model **generalizes structure** to other concepts



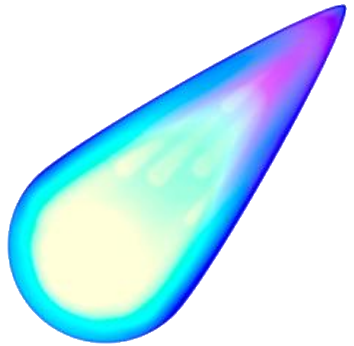
Pre-trained
Language Model

+



Seed Knowledge
Graph Training

=



COMET

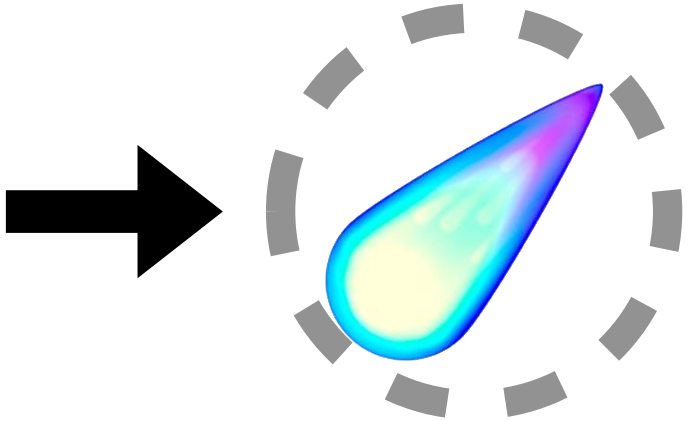
What are the implications of this knowledge representation?

Commonsense Knowledge for any Situation

transformer-style architecture — input format is natural language

- event can be fully parsed

Kai knew that things were getting out of control and managed to keep his temper in check

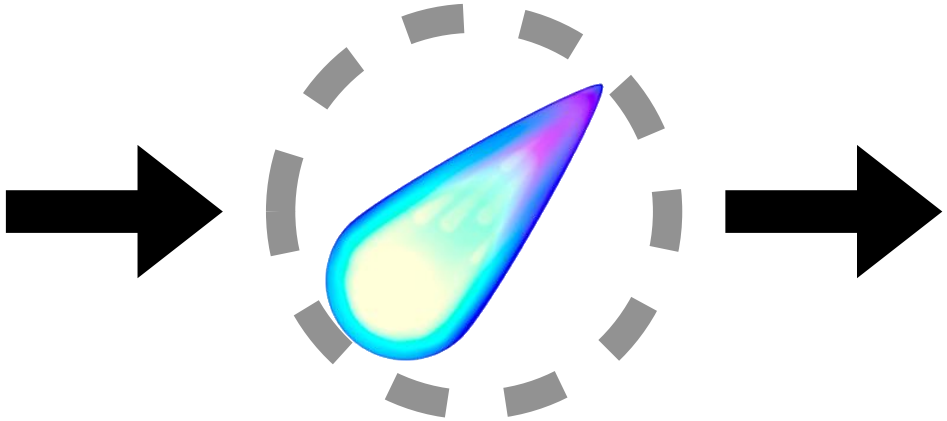


Commonsense Knowledge for any Situation

transformer-style architecture — input format is natural language

- event can be fully parsed
- knowledge generated dynamically from neural knowledge model

Kai knew that things were getting out of control and managed to keep his temper in check



Kai wants to avoid trouble

Kai intends to be calm

Kai stays calm

Kai is viewed as cautious

Symbolic “Wrappers”

The previous example showed how knowledge bases can be used for training, but they can also be used during inference

Helps validate what the NN is doing

VerbNet v3.4

https://verbs.colorado.edu/verbnet/

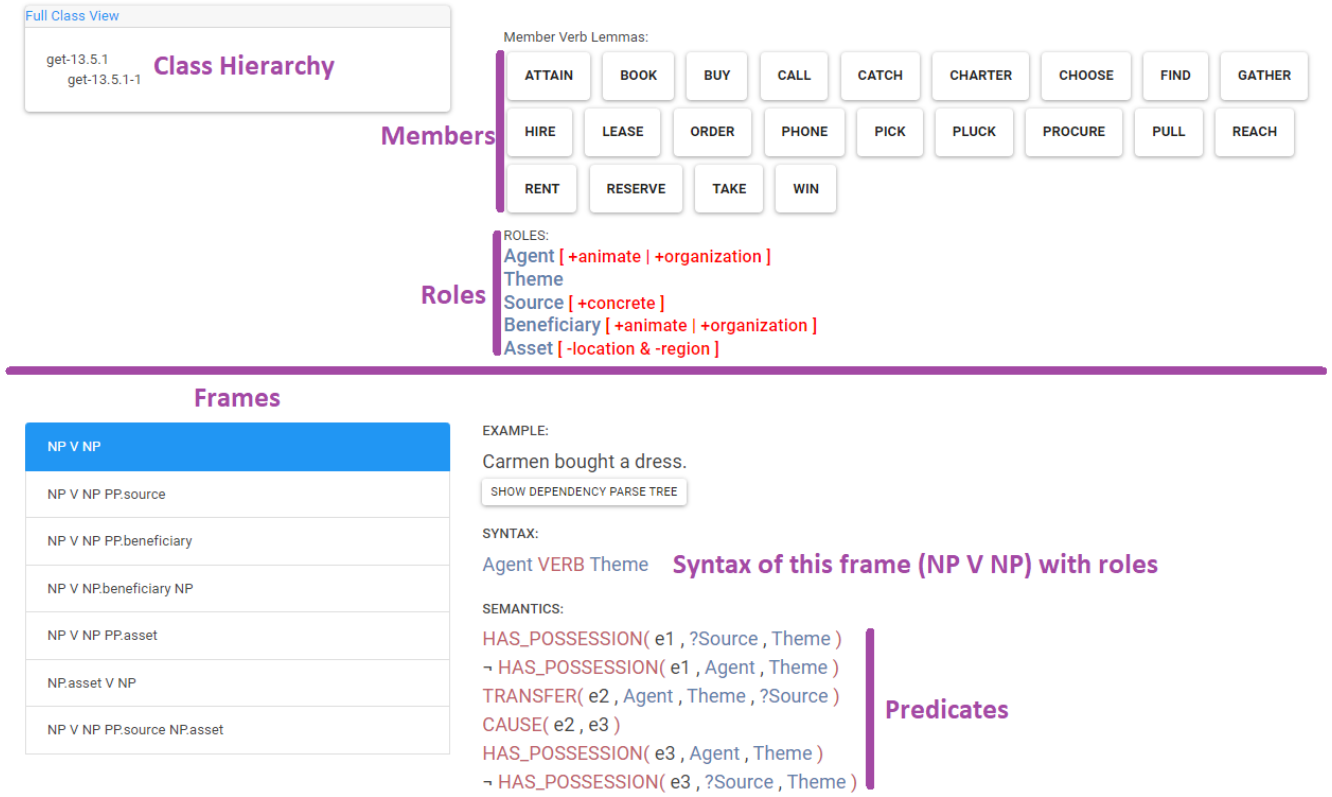
Verb classes based on Beth Levin (1993)

Data Source: hand-crafted

Languages: English

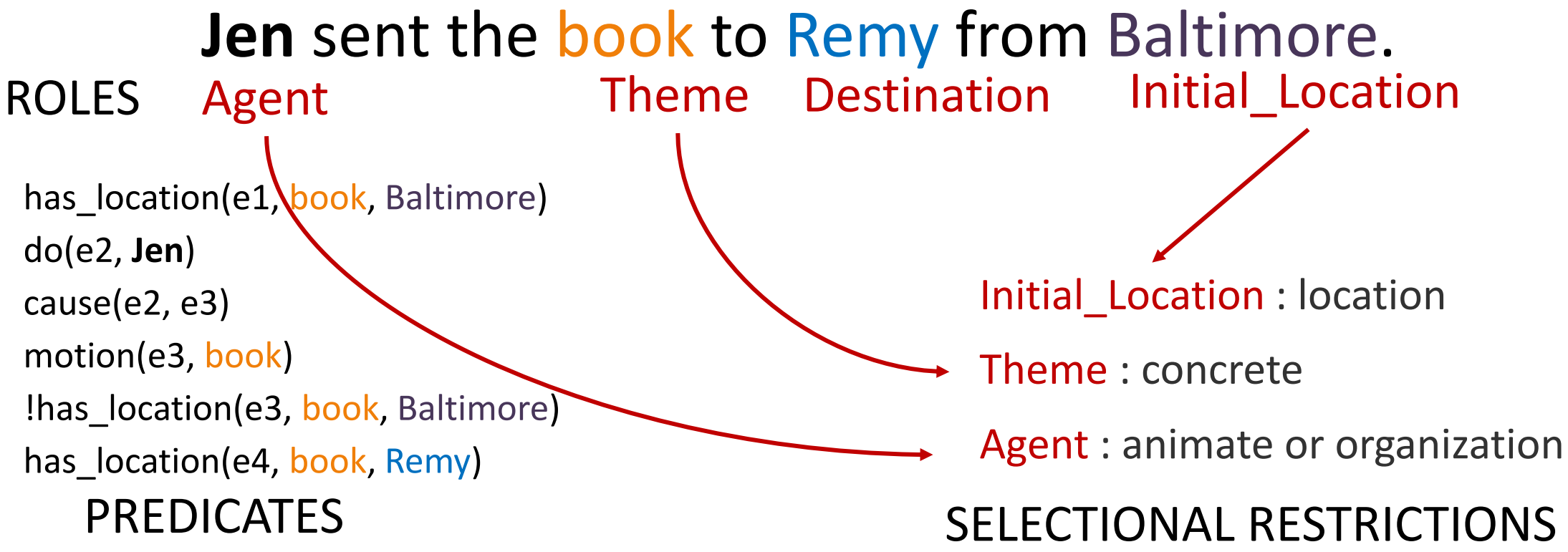
Use: raw data or my code

Demo: https://uvi.colorado.edu/uvi_search



K. Kipper Schuler, “VerbNet: A Broad-Coverage, Comprehensive Verb Lexicon,” University of Pennsylvania, 2005.
Levin, B. (1993) “English Verb Classes and Alternations: A Preliminary Investigation”, University of Chicago Press, Chicago, IL.

Using VerbNet



Pre-Conditions and Effects

Jen sent the book to Remy from Baltimore.

~~do(e2, **Jen**)~~
~~cause(e2, e3)~~
~~motion(e3, book)~~

Pre-Conditions

has_location(e1, book, Baltimore)

Baltimore : location

book : concrete

Jen : animate or organization

Effects

!has_location(e3, book, Baltimore)

has_location(e4, book, Remy)

Pre-Conditions and Effects

Jen sent the book to Remy from Baltimore.



has_location(book, Baltimore)
Baltimore : location
book : concrete
Jen : animate or organization

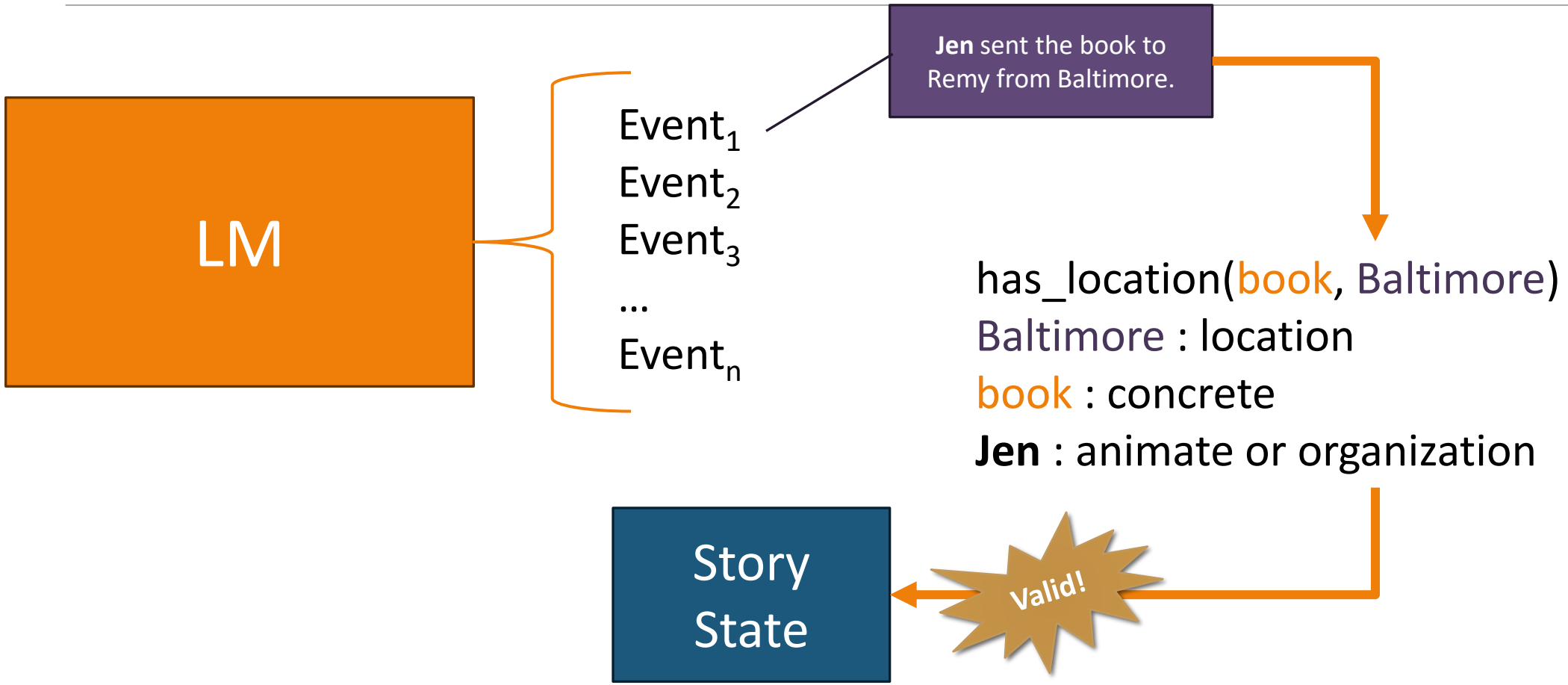
!has_location(book, Baltimore)
has_location(book, Remy)

Resulting State Representation

Jen sent the book to Remy from Baltimore.

Baltimore : location
book : concrete
Jen : animate or organization
!has_location(book, Baltimore)
has_location(book, Remy)

How might a neural network fit in here?



Think-Pair-Share

1. Why might knowledge bases still be useful with today's zero-shot LMs?
2. What are some ways you would integrate a knowledge base into a modern LM?

World Modeling, State Tracking, & Experience Managers

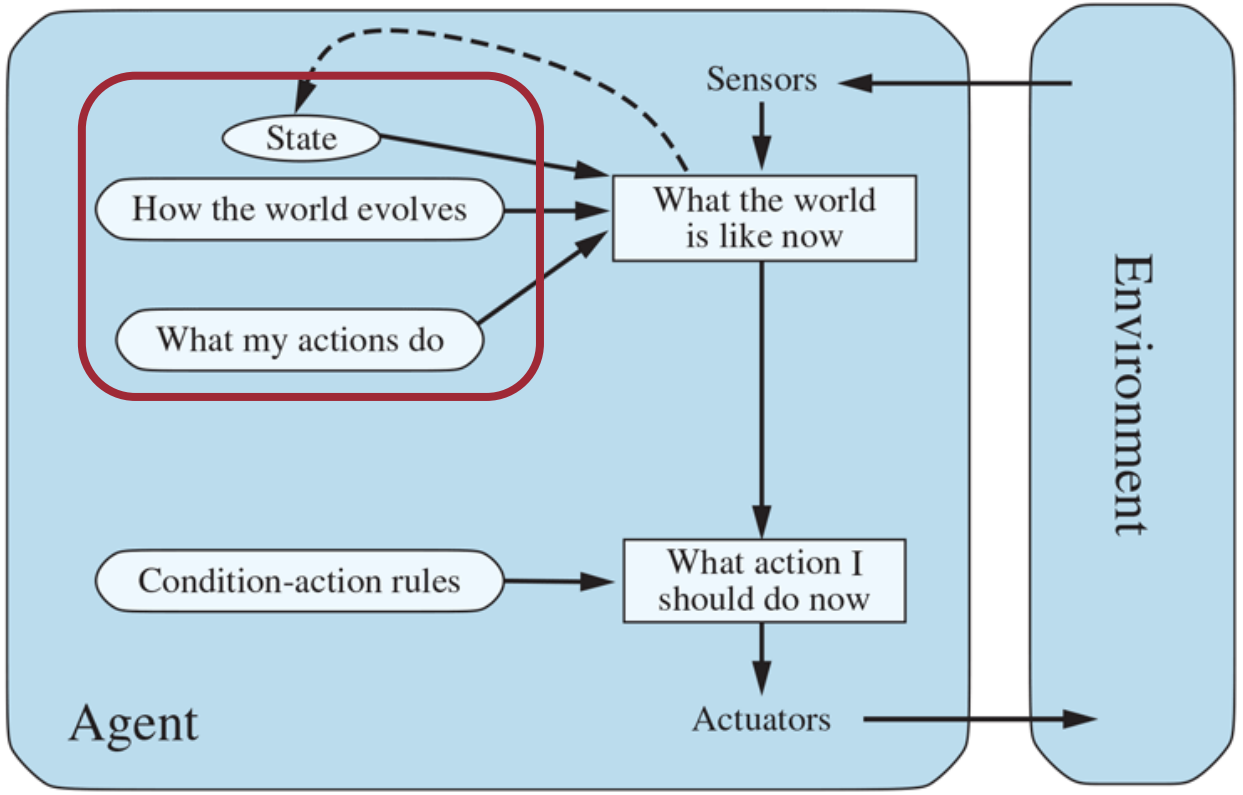
Review: Model-based Reflex Agent

Attempt to model the state of the world, decide action based on modeled ("remembered") state

Example:

Observe that at a certain time, people are more likely to show up in A

Move to location based on time of day



Artificial Intelligence: A Modern Approach, Pg. 52

World Modeling

World modeling is enabling AI agents to have a “mental model” of their environment

May or may not be accurate to the real environment (whether it’s a fictional world, simulation, or the real world)

- Due to sensor error, incorrect parsing, perspective, etc.

The world model is usually built up over time, but may include knowledge bases to bootstrap the agent’s understanding

World Modeling vs State Tracking

State tracking is a term used when describing dialog systems (will discuss more in Module 5)

- Describes systems that find the state of the conversation based on pragmatic information (the linguistic study of discourse)
- Has historically consisted of keeping track of "slots"

Slot	Type	Question
ORIGIN	city	What city are you leaving from?
DEST	city	Where are you going?
DEP DATE	date	What day would you like to leave?
DEP TIME	time	What time would you like to leave?
AIRLINE	line	What is your preferred airline?

Experience Managers

An AI “agent that manipulates a virtual world to coerce a participant’s experience to conform to a set of provided properties”[1]

Usually involves a human-in-the-loop, but the concept can be extended to agents controlling other agents

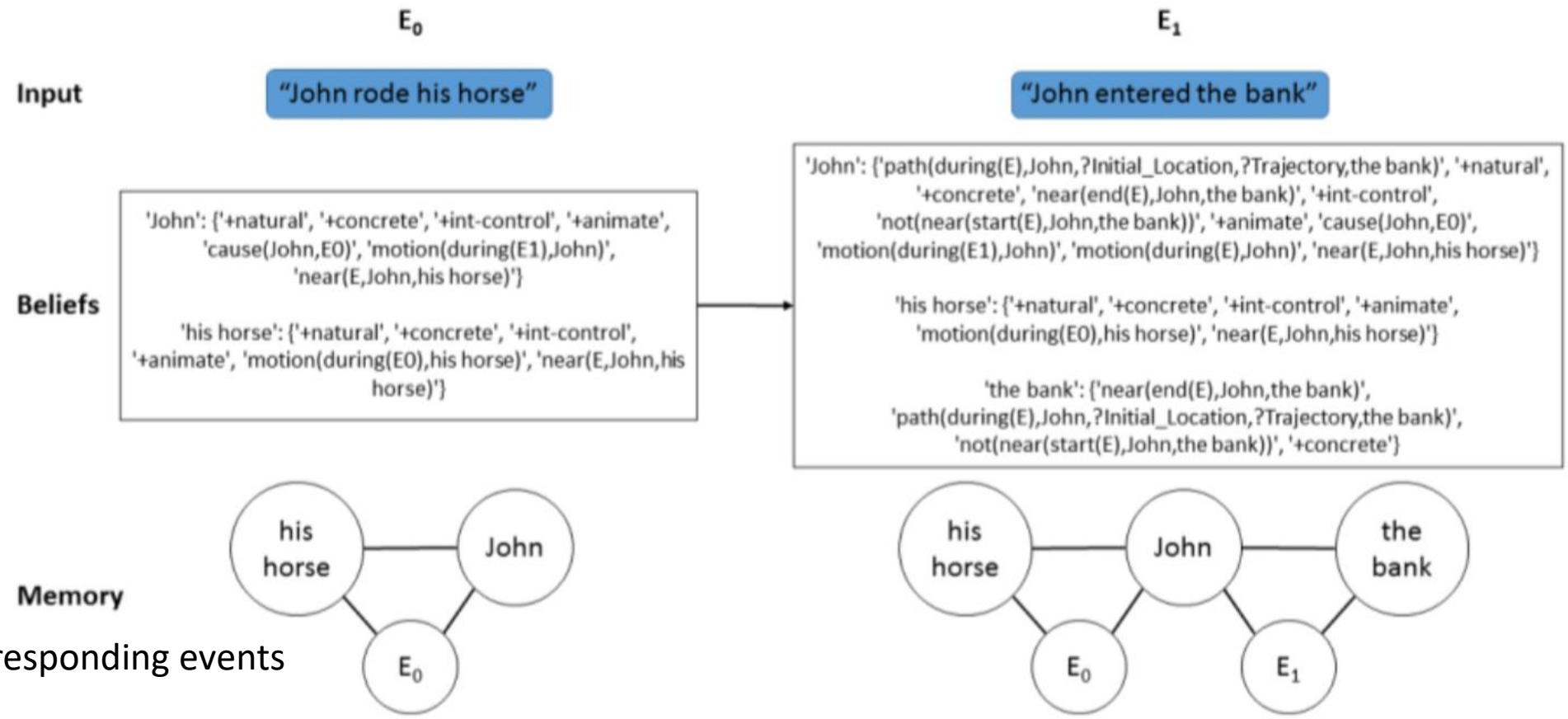
- Humans were “controlled” in the past because AI systems were so rigid

The agent needs to track the state of the world in order to understand where everything is and how to manipulate it

[1] Riedl, Mark O., et al. "Dynamic experience management in virtual worlds for entertainment, education, and training." *International Transactions on Systems Science and Applications, Special Issue on Agent Based Systems for Human Learning* 4.2 (2008): 23-42.

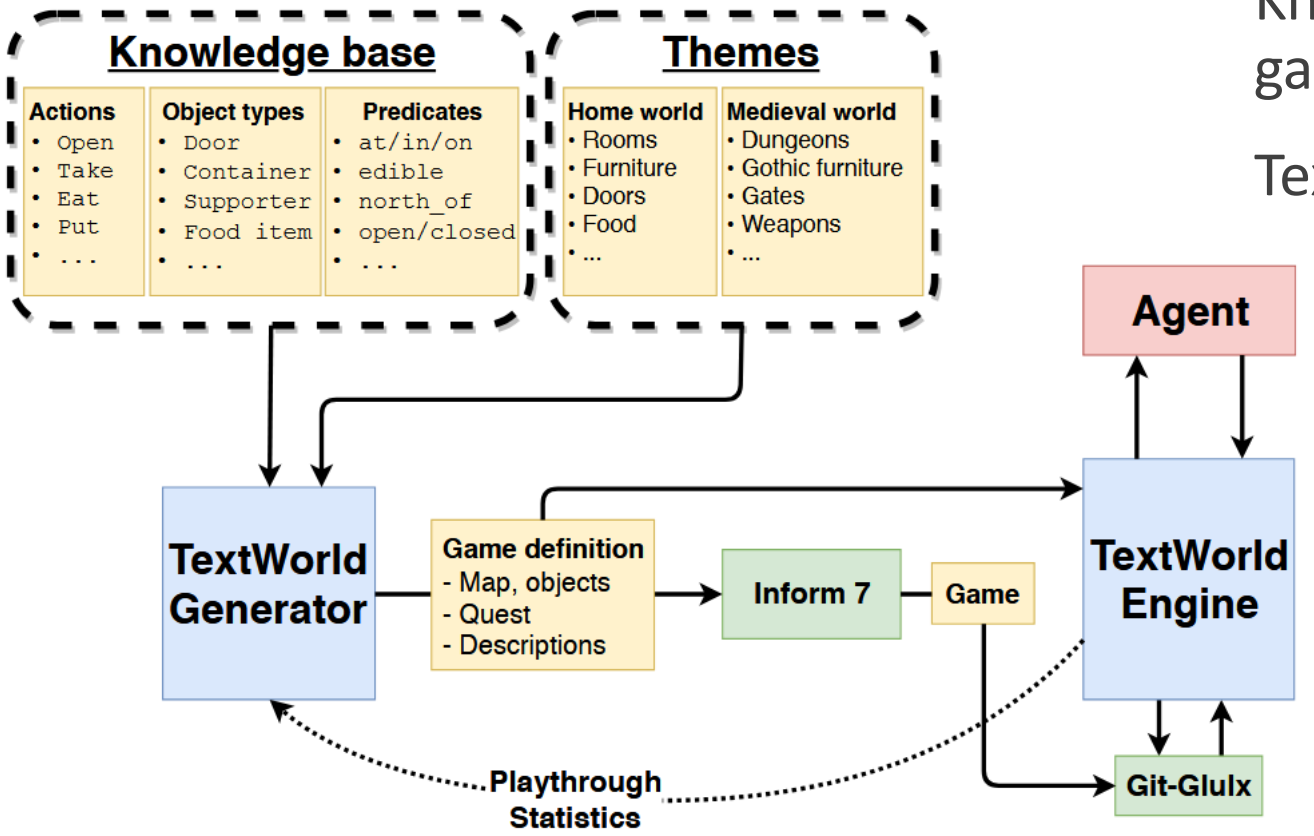
Examples of World Modeling

World Modeling Using VerbNet



Entities and corresponding events

TextWorld



Knowledge-guided building of text games

Text generation using a CFG

Côté, MA. *et al.* (2019). TextWorld: A Learning Environment for Text-Based Games. In: Cazenave, T., Saffidine, A., Sturtevant, N. (eds) Computer Games. CGW 2018. Communications in Computer and Information Science, vol 1017. Springer, Cham. https://doi.org/10.1007/978-3-030-24337-1_3

StoRM

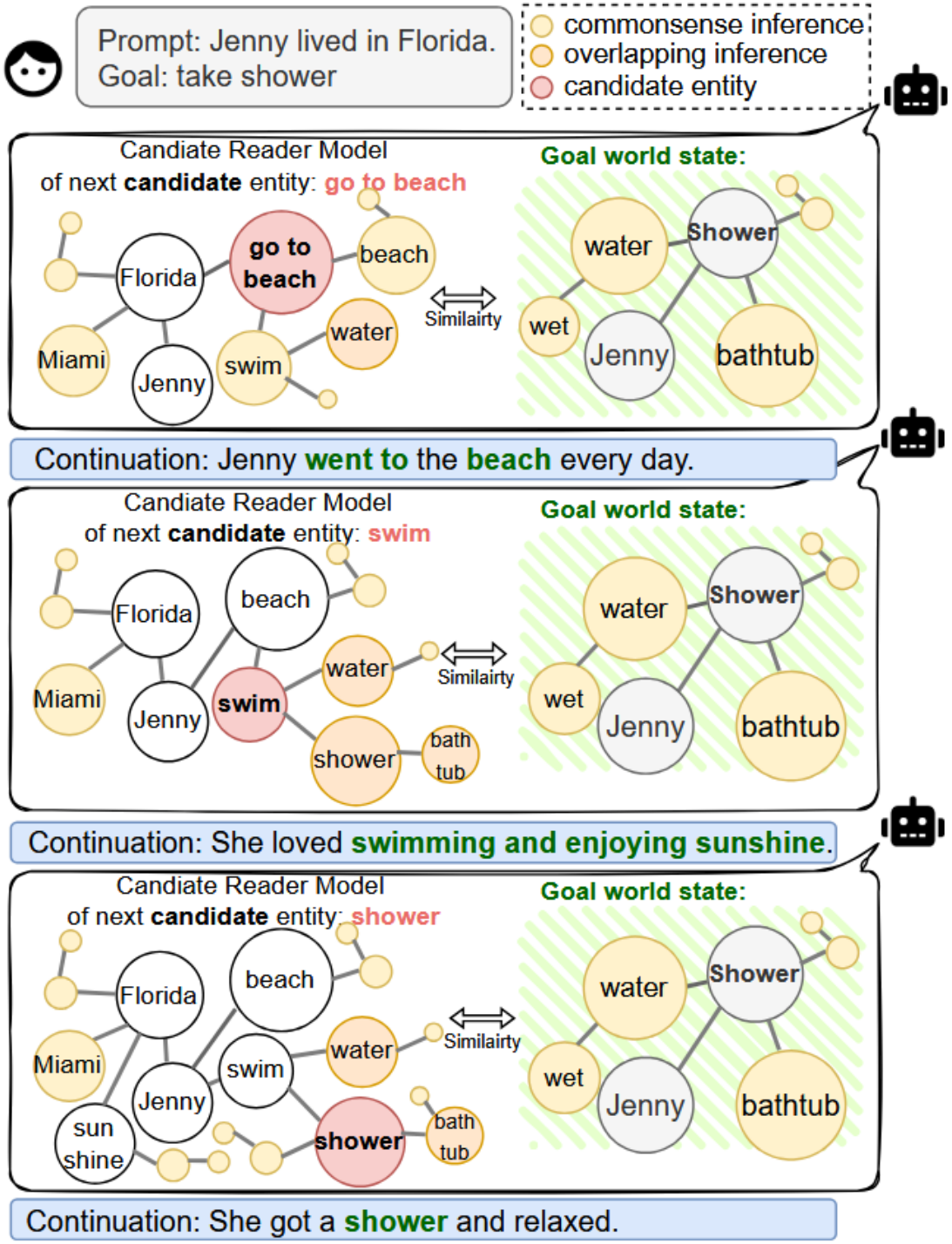
Reader model and goal world states compared

Tuples extracted from sentences are turned into knowledge graphs using semantic role labeling (i.e., filling in roles of frames)

KG expanded with COMET2020

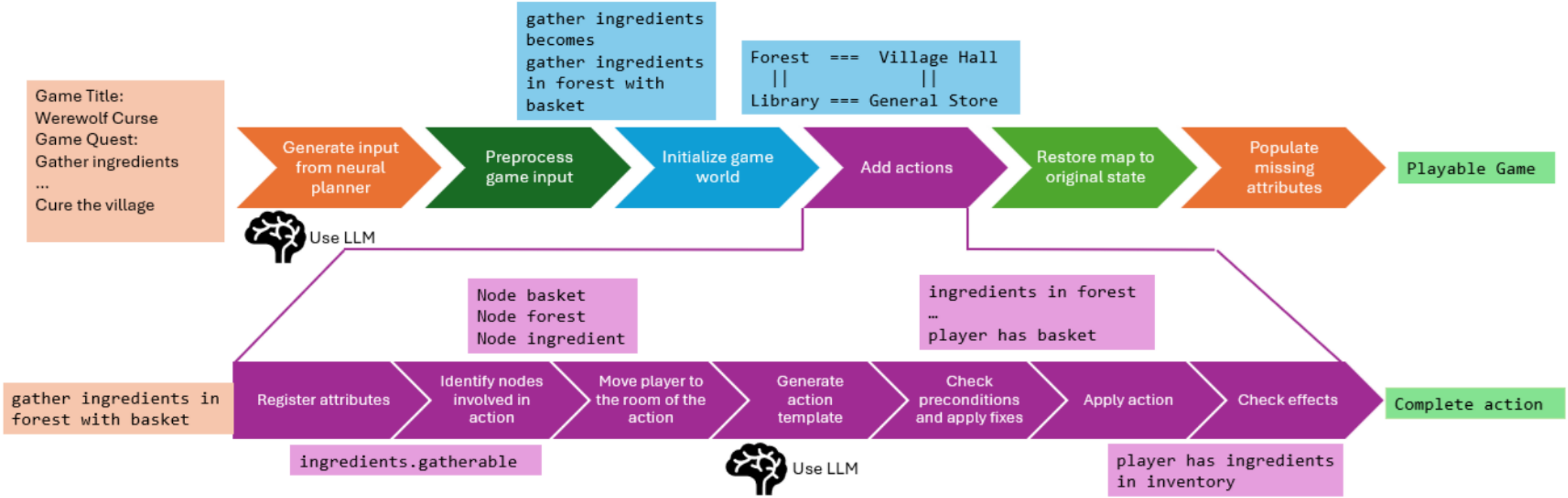
Templates were filled in with RoBERTa (infilling)

- `[subject] <mask> <mask> swim <mask>`



Guiding Neural Story Generation with Reader Models (Peng et al., Findings 2022)

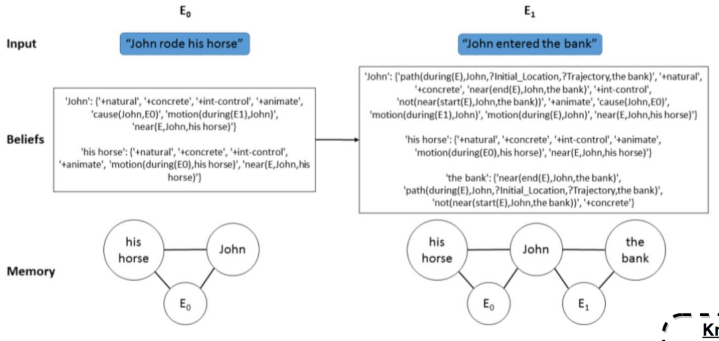
Story2Game


gather ingredients becomes gather ingredients in forest with basket
Forest === Village Hall
|| ||
Library === General Store
Game Title:
Werewolf Curse
Game Quest:
Gather ingredients
...
Cure the village
Generate input from neural planner
Preprocess game input
Initialize game world
Add actions
Restore map to original state
Populate missing attributes
Playable Game
Use LLM
Node basket
Node forest
Node ingredient
ingredients in forest
…
player has basket
gather ingredients in forest with basket
Register attributes
Identify nodes involved in action
Move player to the room of the action
Generate action template
Check preconditions and apply fixes
Apply action
Check effects
Complete action
ingredients.gatherable
Use LLM
player has ingredients in inventory

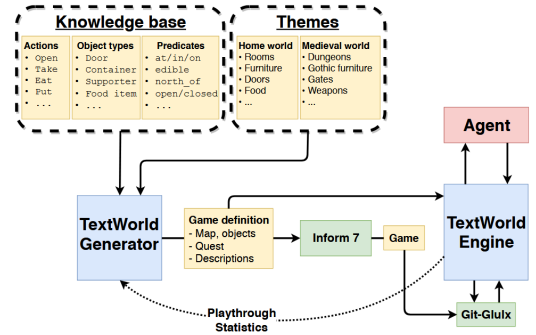
Think-Pair-Share

How has the way we track world states or modeled worlds changed over the years?

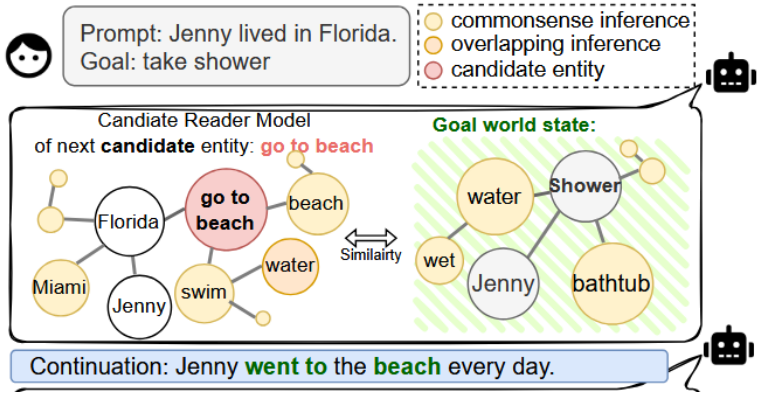
What has stayed the same?



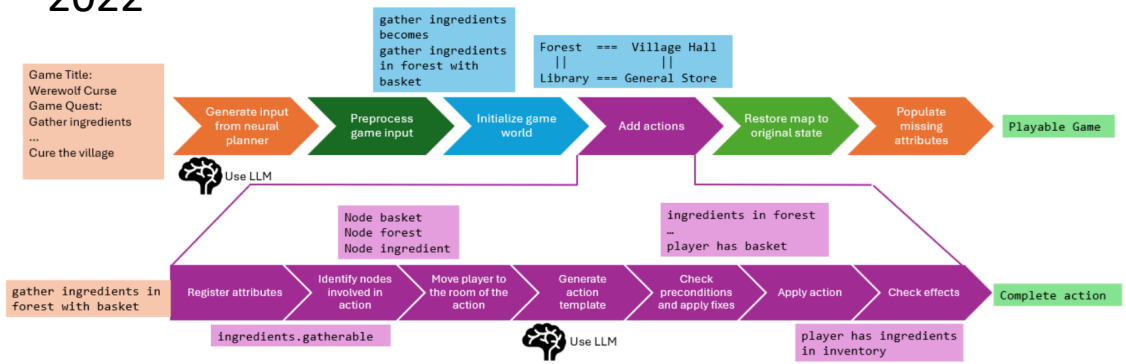
2016



2018



2022



2025